accumulatedMutations_HSPC

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Contents

1 Env

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure5/")
```

2 Libraries

```r
library(FSA)
## ## FSA v0.10.0. See citation('FSA') if used in publication.
## ## Run fishR() for related website and fishR('IFAR') for related book.
library(ineq)
library(vegan)
## Loading required package: permute
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##     filter, lag
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
library(tidyr)
library(ggpubr)
## Loading required package: ggplot2
## Registered S3 methods overwritten by 'car':
##   method       from
##   hist.boot    FSA
##   confint.boot FSA
library(Seurat)
## Loading required package: SeuratObject
## Loading required package: sp
## 'SeuratObject' was built under R 4.5.0 but the current version is
## 4.5.1; it is recomended that you reinstall 'SeuratObject' as the ABI
## for R may have changed
##
## Attaching package: 'SeuratObject'
## The following objects are masked from 'package:base':
##
##     intersect, t
library(igraph)
##
## Attaching package: 'igraph'
## The following object is masked from 'package:Seurat':
##
##     components
## The following object is masked from 'package:tidyr':
##
##     crossing
## The following objects are masked from 'package:dplyr':
##
##     as_data_frame, groups, union
## The following object is masked from 'package:vegan':
##
##     diversity
## The following object is masked from 'package:permute':
##
##     permute
```

```r
## The following objects are masked from 'package:stats':
##
##     decompose, spectrum
## The following object is masked from 'package:base':
##
##     union
library(ggraph)
##
## Attaching package: 'ggraph'
## The following object is masked from 'package:sp':
##
##     geometry
library(webshot)
library(ggplot2)
library(ggpmisc)
## Loading required package: ggpp
## Registered S3 methods overwritten by 'ggpp':
##   method                  from
##   heightDetails.titleGrob ggplot2
##   widthDetails.titleGrob  ggplot2
##
## Attaching package: 'ggpp'
## The following objects are masked from 'package:ggpubr':
##
##     as_npc, as_npcx, as_npcy
## The following object is masked from 'package:ggplot2':
##
##     annotate
library(stringr)
library(corrplot)
## corrplot 0.95 loaded
library(pheatmap)
library(reshape2)
##
## Attaching package: 'reshape2'
## The following object is masked from 'package:tidyr':
##
##     smiths
library(tidygraph)
##
## Attaching package: 'tidygraph'
## The following object is masked from 'package:igraph':
##
##     groups
## The following object is masked from 'package:stats':
##
##     filter
library(networkD3)
##
## Attaching package: 'networkD3'
## The following object is masked from 'package:Seurat':
##
##     JS
```

```
## The following object is masked from 'package:SeuratObject':
##
##     JS
library(tidyverse)
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x lubridate::%--%()       masks igraph::%--%()
## x ggpp::annotate()        masks ggplot2::annotate()
## x tibble::as_data_frame() masks igraph::as_data_frame(), dplyr::as_data_frame()
## x purrr::compose()        masks igraph::compose()
## x igraph::crossing()      masks tidyr::crossing()
## x tidygraph::filter()     masks dplyr::filter(), stats::filter()
## x dplyr::lag()            masks stats::lag()
## x purrr::simplify()       masks igraph::simplify()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
library(patchwork)
```

3 Functions

```
retriveCellsByLocation <- function(CSM, locList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[, c("CellTag",locList)]
  data_MutPerCell_CSM_tmp <- data.frame(
    CellTag = rownames(CSM_tmp),
    MutNum = rowSums(CSM_tmp == "Mut"),
    WTNum = rowSums(CSM_tmp == "WT")
  )
  cells_Total <- data_MutPerCell_CSM_tmp[rowSums(data_MutPerCell_CSM_tmp[,c("MutNum", "WTNum")]) > 0, ]
  cells_WT <-    data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[,"WTNum"] > 0,]$CellTag
  cells_Mut <-   data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[,"MutNum"] > 0,]$CellTag
  cells <- list(Total = cells_Total, Mut = cells_Mut, WT = cells_WT)
  return(cells)
}

retriveLocsByCells <- function(CSM, cellList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[CSM$CellTag %in% cellList,]
  data_CellPerMut_CSM_tmp <- data.frame(
    Location = colnames(CSM_tmp),
    CellMutNum = colSums(CSM_tmp == "Mut"),
    CellWTNum = colSums(CSM_tmp == "WT")
  )
  pos_Total <- data_CellPerMut_CSM_tmp[rowSums(data_CellPerMut_CSM_tmp[, c("CellMutNum", "CellWTNum")])
  pos_Mut <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellMutNum"] > 0 ,]$Location
  pos_WT <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellWTNum"] > 0 ,]$Location
  pos <- list(Total = pos_Total, Mut = pos_Mut, WT = pos_WT)
  return(pos)
}
```

```r
cal_MultiSpe <- function(matrixData) {

  res <- data.frame(
    Shannon = vegan::diversity(matrixData, index = "shannon"),
    Simpson = vegan::diversity(matrixData, index = "simpson"),
    Gini = apply(matrixData, 1, function(x) ineq(x + 0.00001, type = "Gini")),
    sumMut = rowSums(matrixData),
    Tau = NA
  )

  # Tau = sum(1 - (x / max(x))) / (n - 1)
  res$Tau <- apply(matrixData, 1, function(x) {
    if (all(x == 0)) return(NA)
    x <- x / max(x)
    sum(1 - x) / (length(x) - 1)
  })

  rownames(res) <- rownames(matrixData)
  return(res)
}

scale01 <- function(x) (x - min(x)) / (max(x) - min(x))

cal_scMutScore <- function(data_cell_mut, data_mut_time){
  # data_cell_mut [cell * muts]
  # data_mut_time [muts * tissues/celltype/days]
  # data_cell_time_matrix
  replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

  data_cell_mut_num <- as.data.frame(lapply(data_cell_mut, function(x) {
    ifelse(x %in% names(replacement_map), replacement_map[x], x)
  }))

  rownames(data_cell_mut_num) <- rownames(data_cell_mut)
  print(dim(data_cell_mut_num))
  print(dim(data_mut_time))

  data_cell_time_matrix <- as.matrix(data_cell_mut_num) %*% as.matrix(data_mut_time)
  rownames(data_cell_time_matrix) <- rownames(data_cell_mut)
  colnames(data_cell_time_matrix) <- colnames(data_mut_time)

  muts_total <- unique(rownames(data_mut_time))

  # get different tissues/time/celltype names
  col_names_data <- colnames(data_cell_time_matrix)

  col_lenghth <- length(col_names_data)

  col_coefficient_fix <- setNames(
    rep(1/col_lenghth, col_lenghth),
    col_names_data
  )
```

```
  data_cell_time_matrix <- data_cell_time_matrix[,col_names_data]

  # Gini
  res_index <- cal_MultiSpe(data_cell_time_matrix)

  res_diversity <- as.data.frame((data_cell_time_matrix > 1) * rep(col_coefficient_fix[colnames(data_cel
  res_diversity$Diversity <- rowSums(res_diversity)

  res_diversity$Diversity[is.na(res_diversity$Diversity)] <- 0.00001

  res_index$Gini[is.na(res_index$Gini)] <- 0.00001
  res_index$Tau[is.na(res_index$Tau)] <- 0.00001

  res_index$GiniTau <- res_index$Gini * res_index$Tau

  res_index$scMutScore<- 2 / (
    (1 / (res_index$GiniTau + 0.00001)) +
    (1 / (res_diversity$Diversity + 0.00001))
  )

  res_index$Diversity <- res_diversity$Diversity

  res_index$scMutScore <- scales::rescale(res_index$scMutScore, to = c(0, 1))

  res <- list(diversity = res_diversity, score = res_index, matrixD = data_cell_time_matrix)
  return(res)
}
```

4 LoadData

```
CD34_CSM <- readRDS("../Data/CD34/CD34_CSM.rds")
CD34_SSM <- readRDS("../Data/CD34/CD34_SSM.rds")
CD34_Meta <- readRDS("../Data/CD34/CD34_meta.rds")
```

5 Calculationg mutations of each cell and cells of each mutation

```
CD34_MutPerCell <- data.frame(
  CellTag = rownames(CD34_CSM),
  MutNum_somatic = rowSums(CD34_CSM == "Mut"),
  WTNum_somatic = rowSums(CD34_CSM == "WT")
)

CD34_CellPerMut <- data.frame(
  Location = colnames(CD34_CSM),
  CellMutNum = colSums(CD34_CSM == "Mut"),
  CellWTNum = colSums(CD34_CSM == "WT")
)
```

6 calMulti

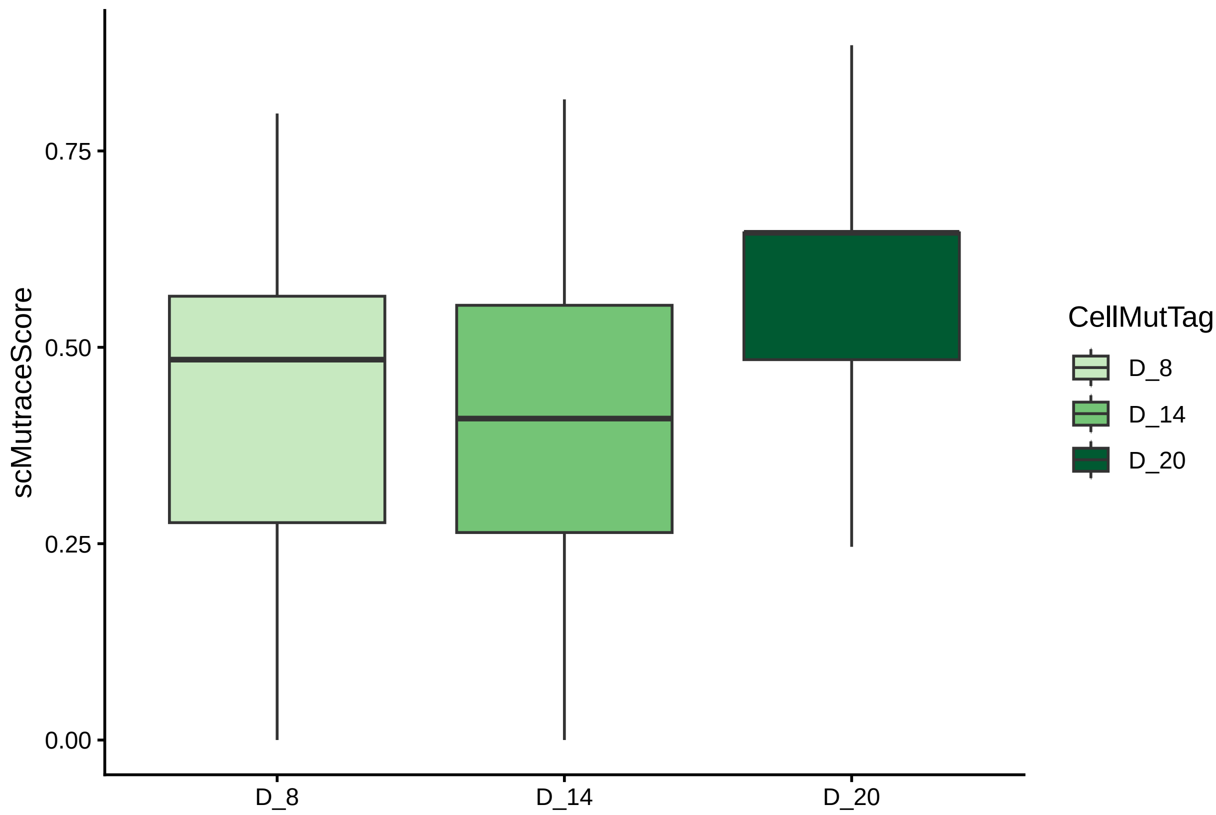
```r
time_types <- c("CD34_800_Day08", "CD34_800_Day14", "CD34_800_Day20")

# Cell_Mut
cells_candidate <- CD34_MutPerCell[CD34_MutPerCell$MutNum_somatic >= 1,]$CellTag
muts_candidate <- CD34_CellPerMut[CD34_CellPerMut$CellMutNum >= 1,]$Location

CD34_cell_mut_raw <- CD34_CSM[cells_candidate, muts_candidate]

# Mut_Time
CD34_mut_time <- as.data.frame(matrix(nrow = length(muts_candidate), ncol = length(time_types)))
rownames(CD34_mut_time) <- muts_candidate
colnames(CD34_mut_time) <- time_types

for (i in 1:nrow(CD34_mut_time)){
  mut_tmp <- rownames(CD34_mut_time)[i]
  cell_tmp <- retriveCellsByLocation(CD34_cell_mut_raw, mut_tmp)
  selected_meta_tags <- CD34_Meta[CD34_Meta$Cellbarcode %in% cell_tmp$Mut,]$Cellbarcode

  CD34_mut_time[i,]$CD34_800_Day08 <- sum(str_count(selected_meta_tags, "CD34_800_Day08"))
  CD34_mut_time[i,]$CD34_800_Day14 <- sum(str_count(selected_meta_tags, "CD34_800_Day14"))
  CD34_mut_time[i,]$CD34_800_Day20 <- sum(str_count(selected_meta_tags, "CD34_800_Day20"))
}
```

7 computescMutScore

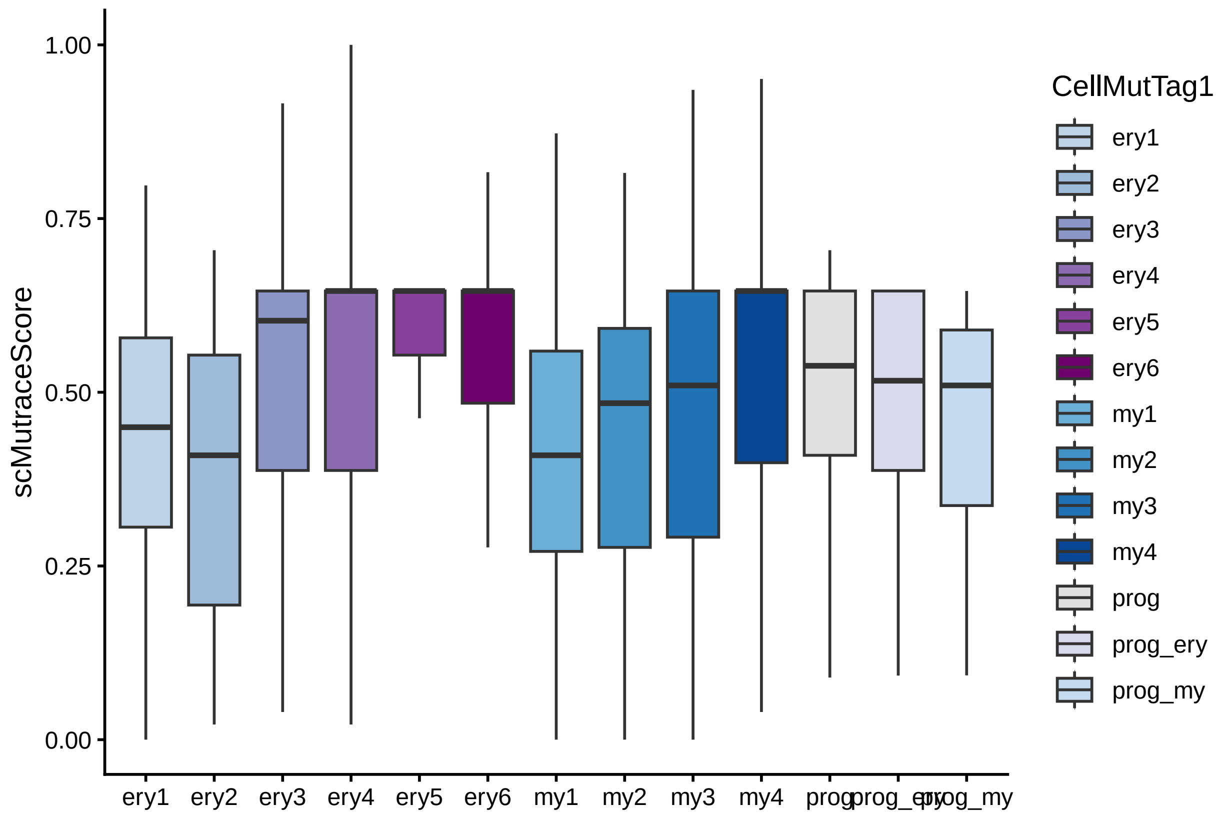
```r
res_index_raw_total <- cal_scMutScore(CD34_cell_mut_raw, CD34_mut_time)
## [1] 1893  478
## [1] 478   3

res_index_raw <- res_index_raw_total$score
res_index_raw$Cellbarcode <- rownames(res_index_raw)

res_matrix <- res_index_raw_total$matrixD

res_index <- merge(res_index_raw, CD34_Meta, by = "Cellbarcode")

selectedGroup <- c("prog", "prog_my", "prog_ery", "my1","my2",'my3','my4', "ery1", "ery2", "ery3", "ery
res_index <- res_index[res_index$Group %in% selectedGroup,]

res_index$CellMutTag <- factor(res_index$CellMutTag, levels = c("D_8", "D_14", "D_20"))
res_index <- res_index[res_index$Diversity > 0, ]
```

8 checkIndex

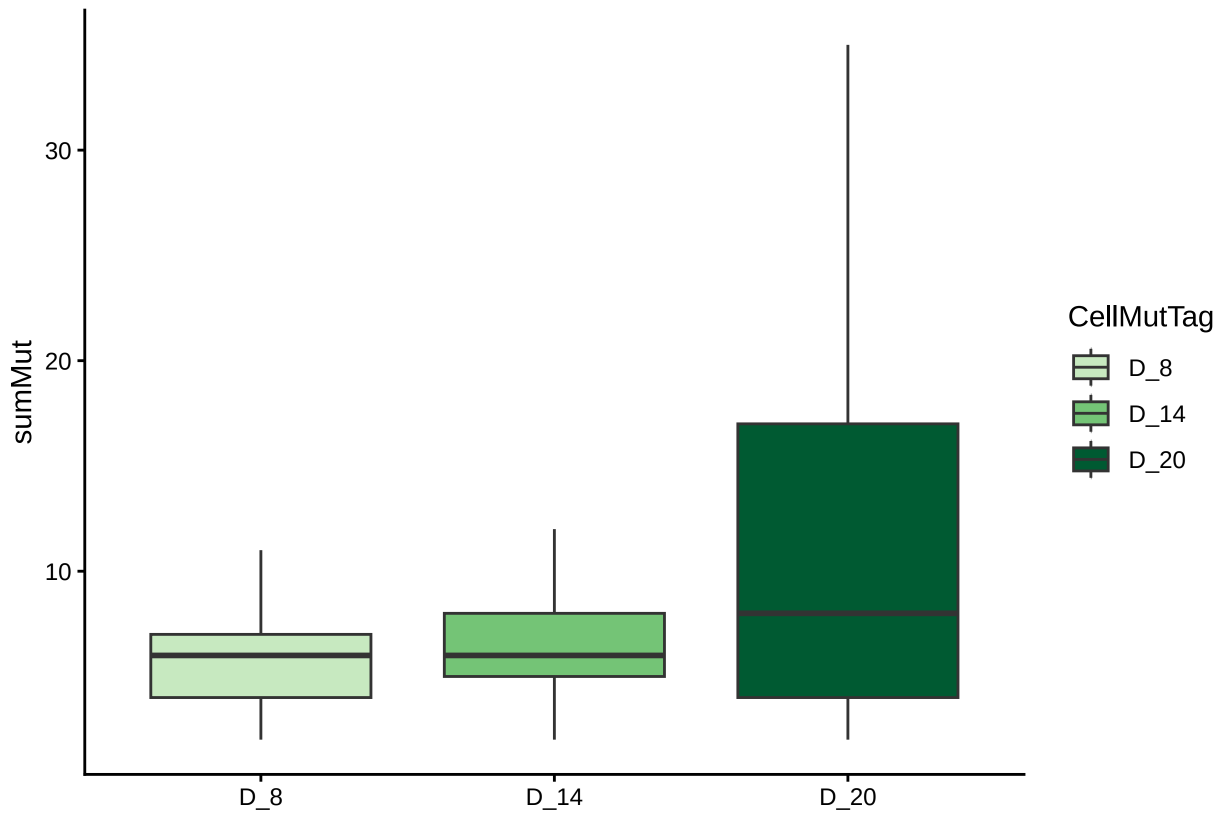
```r
cluster_color <- c(my1 = '#6baed6', my2 = '#4292c6',  my3 = '#2171b5',  my4 = '#084594', prog = '#e0e0e
                   ery1 = "#bfd3e6", ery2 = "#9ebcda", ery3 = "#8c96c6", ery4 = "#8c6bb1", ery5 = "#884
                   prog_ery = "#d8daeb", mix = "#66c2a4",  Prog = '#c6dbef', ery1_2 = "#8c96c6", ery3_4

day_color <- c(D_8 = "#c7e9c0", D_14 = "#74c476", D_20 = "#005a32")

cell_group_color <- c(Prog = "#e5f5f9", Diff = "#66c2a4", Terminal = "#006d2c")

res_index <- res_index[!is.na(res_index$scMutScore),]

ggplot(data = res_index, mapping = aes(x = CellMutTag, y = scMutScore, fill = CellMutTag)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = day_color) +
  theme_classic()
```



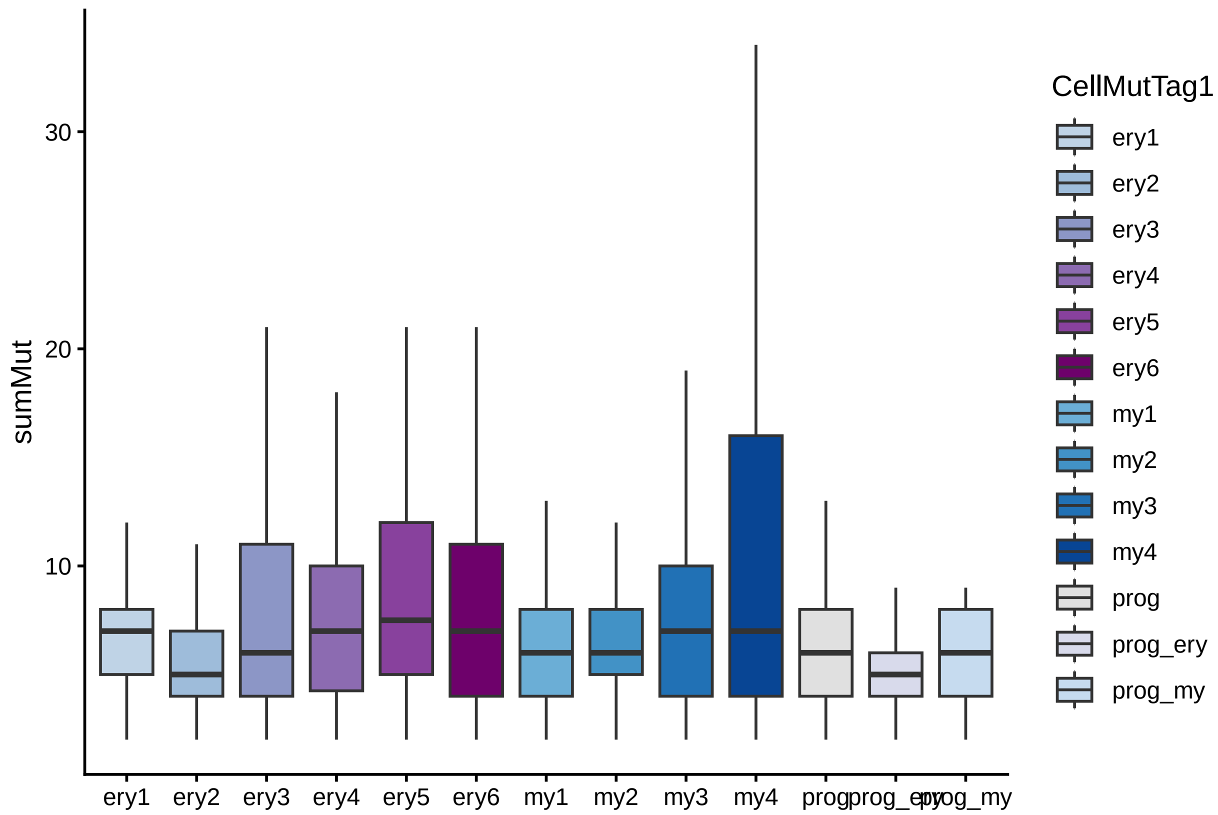

```r
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = scMutScore, fill = CellMutTag1)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()
```



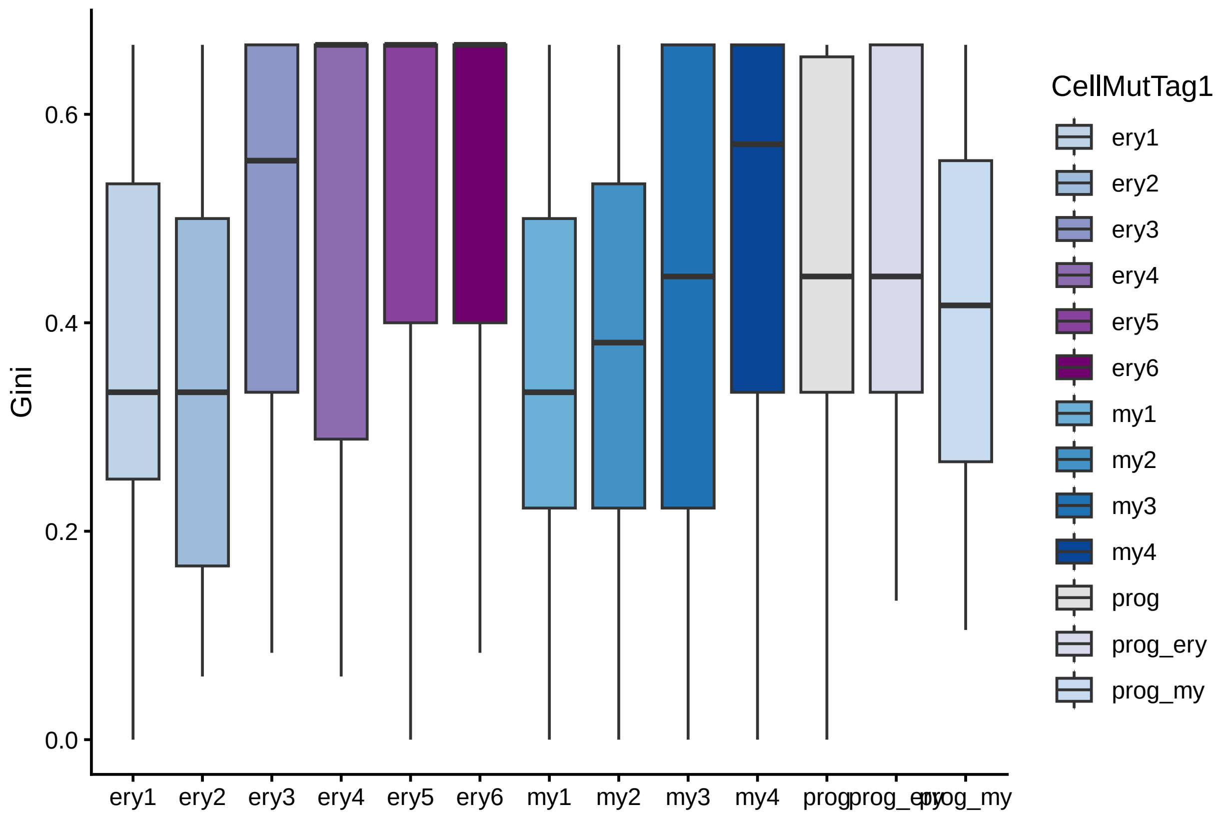

```r
ggplot(data = res_index, mapping = aes(x = CellMutTag, y = sumMut, fill = CellMutTag)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "sumMut") +
  scale_fill_manual(values = day_color) +
  theme_classic()
```



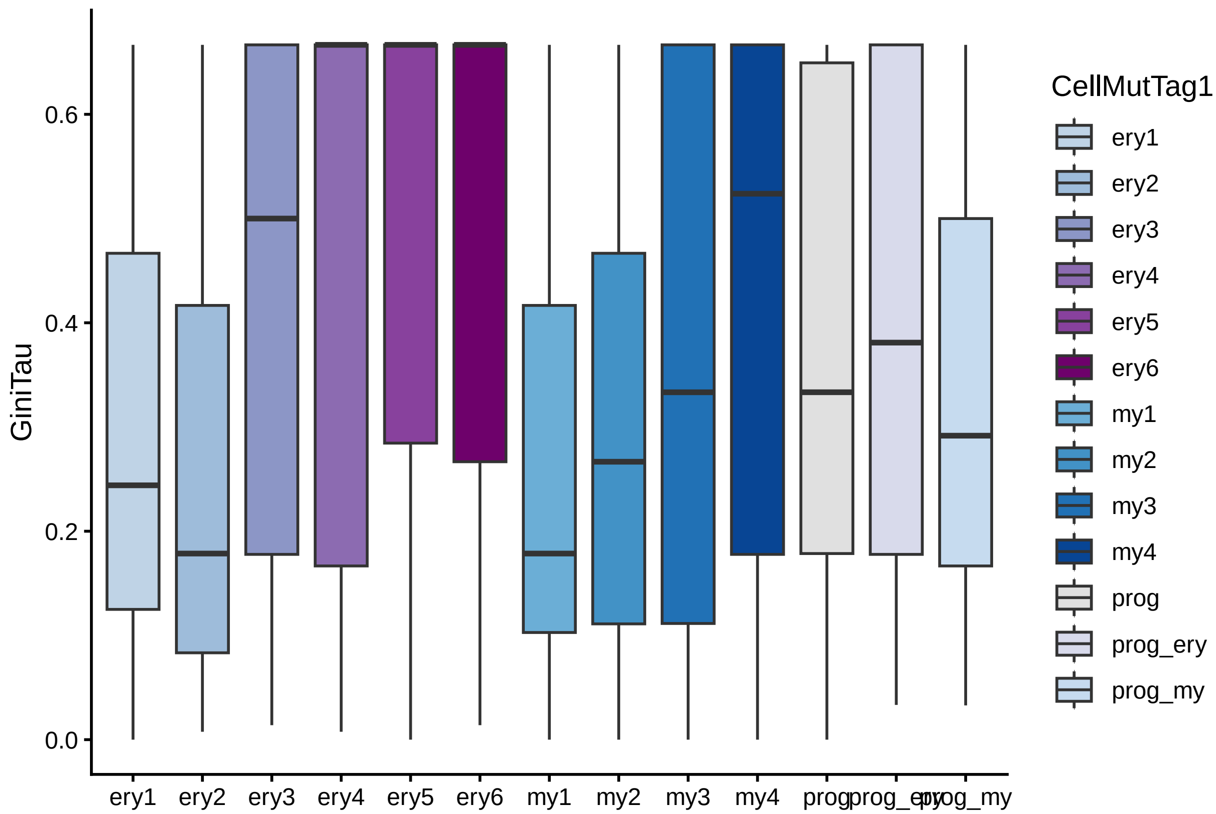

```r
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = sumMut, fill = CellMutTag1)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "sumMut") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()
```



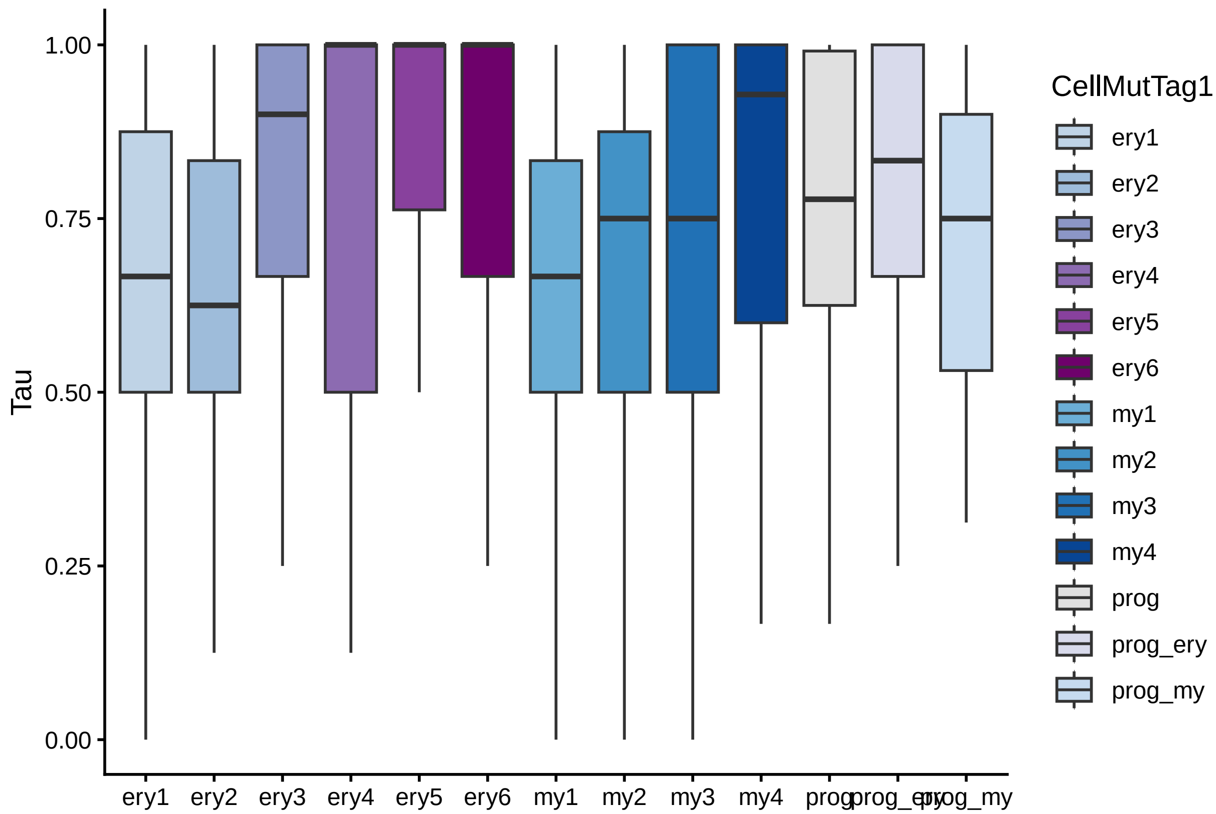

```
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = Gini, fill = CellMutTag1)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "Gini") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()
```



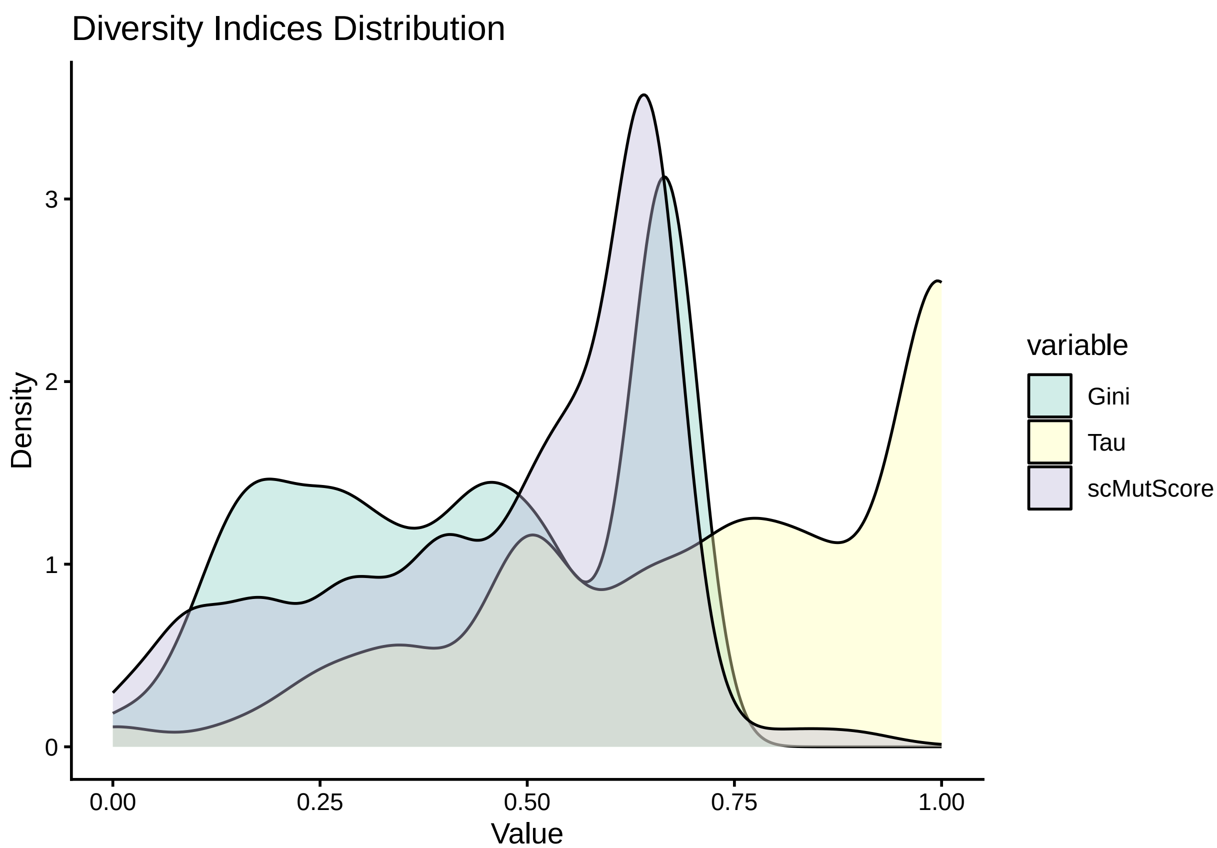

```r
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = GiniTau, fill = CellMutTag1)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "GiniTau") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()
```



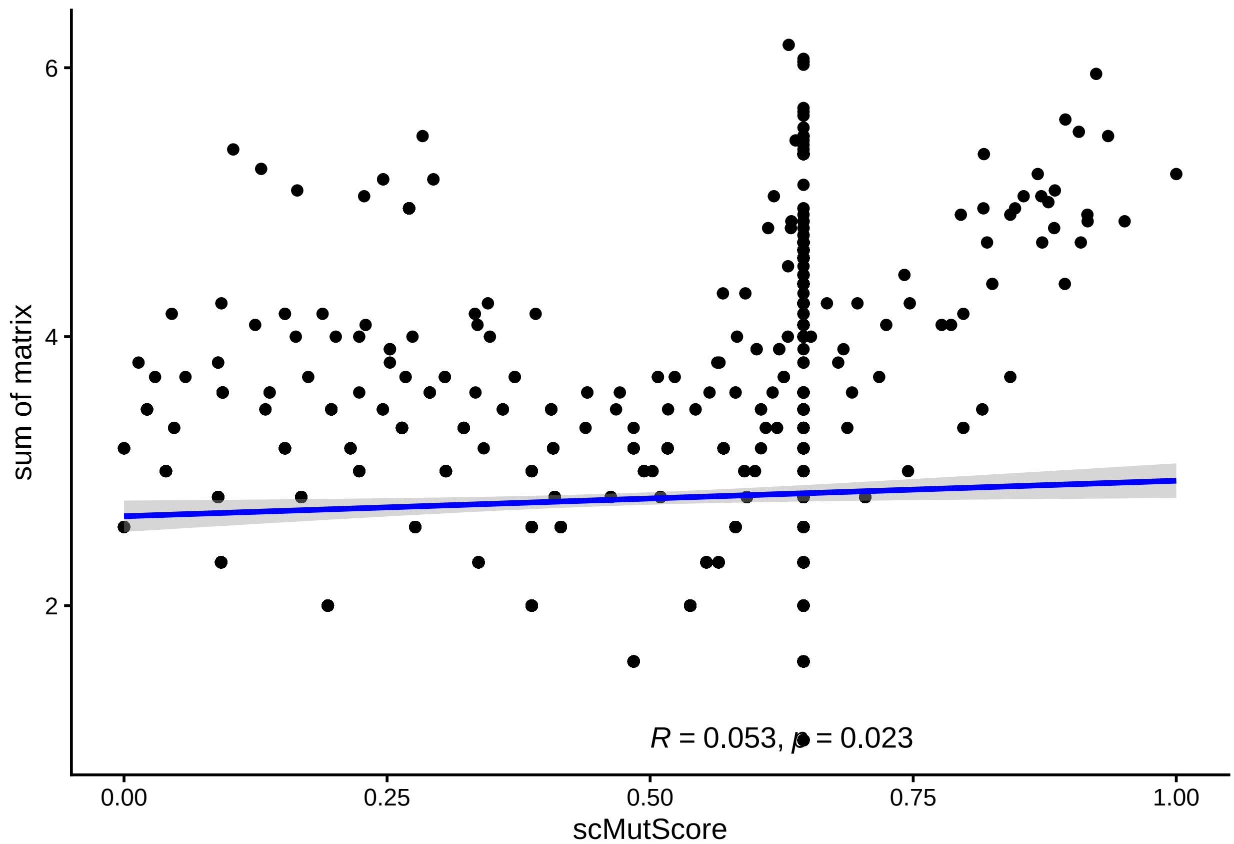

```r
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = Tau, fill = CellMutTag1)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "Tau") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()
```



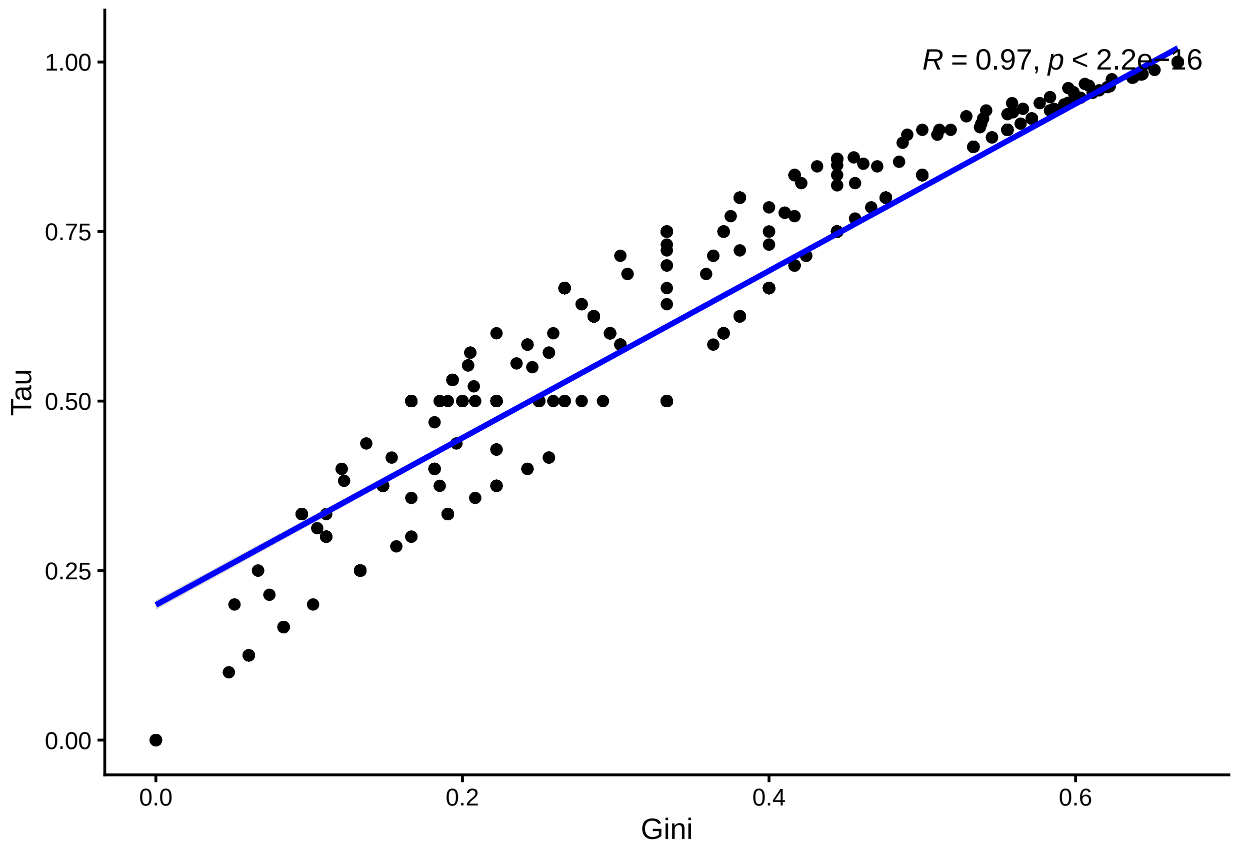

```r
res_index_long <- melt(res_index[,c("Gini","Tau", "scMutScore")])
## No id variables; using all as measure variables

ggplot(res_index_long, aes(x=value, fill=variable)) +
  geom_density(alpha=0.4) +
  theme_classic() +
  labs(x="Value", y="Density", title="Diversity Indices Distribution") +
  scale_fill_brewer(palette="Set3")
```




```
ggplot(res_index, aes(x = scMutScore, y = log2(sumMut))) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  stat_cor(method = "pearson", label.x = 0.5, label.y = 1) +
  labs(x = "scMutScore", y = "sum of matrix") +
  theme_classic()
## `geom_smooth()` using formula = 'y ~ x'
```




```
ggplot(res_index, aes(x = Gini, y = Tau)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  stat_cor(method = "pearson", label.x = 0.5, label.y = 1) +
  labs(x = "Gini", y = "Tau") +
  theme_classic()
## `geom_smooth()` using formula = 'y ~ x'
```



9 scMutrce validation

```
set.seed(2025)

# 1. init
data_check_CM <- CD34_cell_mut_raw
data_check_Meta <- CD34_Meta[CD34_Meta$Cellbarcode %in% rownames(data_check_CM),]
col_check_names <- c("ery1", "ery2", "ery3", "ery4", "ery5", "ery6",  "my1",  "my2",  "my3",  "my4", "p
time_types <- c("CD34_800_Day08", "CD34_800_Day14", "CD34_800_Day20")
cluster_color <- c(my1 = '#6baed6', my2 = '#4292c6',  my3 = '#2171b5',  my4 = '#084594', prog = '#e0e0e
                   ery1 = "#bfd3e6", ery2 = "#9ebcda", ery3 = "#8c96c6", ery4 = "#8c6bb1", ery5 = "#884
                   prog_ery = "#d8daeb", mix = "#66c2a4",  Prog = '#c6dbef', ery1_2 = "#8c96c6", ery3_4

day_color <- c(D_8 = "#c7e9c0", D_14 = "#74c476", D_20 = "#005a32")

cell_num <- 1
mut_num <- 1
mut_loss_ratio <- 0.5
cell_loss_ratio <- 0.5
iteration_times <- 100

# list for save data
data_scMutModel_list <- vector("list", iteration_times)
```

```r
data_scMutScore_list <- vector("list", iteration_times)

num_cells <- nrow(data_check_CM)
num_mutations <- ncol(data_check_CM)
cells_all <- rownames(data_check_CM)
muts_all <- colnames(data_check_CM)

rowname_day <- data.frame(rowname = rownames(data_check_CM), day = sub("_[^_]+$", "", rownames(data_che
num_randomSelected_forEachTime <- ceiling(min(table(rowname_day$day)) * cell_loss_ratio)

cell_day_map <- setNames(rowname_day$day, rowname_day$rowname)

# 2. loop
## ---------------------------------------------------------------
## Precompute balanced sampling plan: every cell will be sampled
## before any cell is repeatedly sampled within each day.
## ---------------------------------------------------------------
cells_by_day <- split(seq_len(nrow(rowname_day)), rowname_day$day)

make_sampling_plan <- function(cells_by_day, n_per_day) {
  lapply(cells_by_day, function(idx) {
    idx_shuffled <- sample(idx)
    split(
      idx_shuffled,
      ceiling(seq_along(idx_shuffled) / n_per_day)
    )
  })
}

sampling_plan <- make_sampling_plan(
  cells_by_day,
  num_randomSelected_forEachTime
)

max_batches <- max(sapply(sampling_plan, length))

# for lineage est
for (iter in seq_len(iteration_times)) {

  # 2.1 randomly loss info
  ## Reshuffle after each full cycle
  if (iter > 1 && (iter - 1) %% max_batches == 0) {
    sampling_plan <- make_sampling_plan(
      cells_by_day,
      num_randomSelected_forEachTime
    )
    max_batches <- max(sapply(sampling_plan, length))
  }

  ## Balanced cell sampling across days
  loss_cell_indices <- unlist(
    lapply(sampling_plan, function(day_batches) {
      batch_id <- ((iter - 1) %% length(day_batches)) + 1
```

```r
      day_batches[[batch_id]]
    }),
    use.names = FALSE
  )

  loss_mut_indices <- sample(seq_len(num_mutations), ceiling(mut_loss_ratio * num_mutations))

  C_partial_missing <- data_check_CM
  # new data
  if(length(loss_cell_indices) > 0 && length(loss_mut_indices) > 0){
    C_partial_missing[loss_cell_indices, loss_mut_indices] <- "NoCall"
  }

  # 2.2 count cell/mut number
  MutPerCell <- rowSums(C_partial_missing == "Mut")
  CellPerMut <- colSums(C_partial_missing == "Mut")

  cells_candidate <- cells_all[MutPerCell >= cell_num & cells_all %in% data_check_Meta$Cellbarcode]
  cells_candidate_group <- data_check_Meta[cells_candidate, "Cellbarcode"]
  muts_candidate <- muts_all[CellPerMut >= mut_num]

  data_cell_mut_raw_check <- C_partial_missing[cells_candidate, muts_candidate, drop = TRUE]
  tab <- table(sub(".*_(Day[0-9]+)_.*", "\\1", rownames(data_cell_mut_raw_check)))
  message(paste0("Iteration: ", iter, " loss_cell_indices: ", length(loss_cell_indices), " loss_mut_ind
  message(paste0("Iteration: ", iter, " cells_candidate: ", length(cells_candidate), " muts_candidate:
  message("Cell counts per day:")
  message(paste(names(tab), tab, sep = ": ", collapse = " | "))

  # 2.3 Mut × Time
  # init Mut × Time
  M_partial_missing <- matrix(0, nrow = length(muts_candidate), ncol = length(time_types),
                              dimnames = list(muts_candidate, time_types))

  for (mut in muts_candidate) {
    cells_with_mut <- rownames(data_cell_mut_raw_check)[data_cell_mut_raw_check[, mut] == "Mut"]
    cells_with_mut_clean <- sub("\\.[0-9]+$", "", cells_with_mut)
    days <- cell_day_map[cells_with_mut_clean]
    M_partial_missing[mut, ] <- table(factor(days, levels = time_types))
  }

  # 2.4 normalized Mut × Time
  M_partial_missing_norm <- M_partial_missing

  # 2.5 scMutScore
  res_index_tmp <- cal_scMutScore(data_cell_mut_raw_check, M_partial_missing_norm)$score
  res_index_tmp$iteration <- paste0("iter", iter)
  res_index_tmp <- res_index_tmp[res_index_tmp$Diversity > 0, ]
  res_index_tmp$Cellbarcode <- rownames(res_index_tmp)
  rownames(res_index_tmp) <- NULL
  data_scMutModel_list[[iter]] <- res_index_tmp
}
## Iteration: 1 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 1 cells_candidate: 1543 muts_candidate: 465
```

```
## Cell counts per day:
## Day08: 485 | Day14: 339 | Day20: 719
## [1] 1543  465
## [1] 465   3
## Iteration: 2 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 2 cells_candidate: 1566 muts_candidate: 466
## Cell counts per day:
## Day08: 494 | Day14: 352 | Day20: 720
## [1] 1566  466
## [1] 466   3
## Iteration: 3 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 3 cells_candidate: 1630 muts_candidate: 467
## Cell counts per day:
## Day08: 533 | Day14: 367 | Day20: 730
## [1] 1630  467
## [1] 467   3
## Iteration: 4 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 4 cells_candidate: 1582 muts_candidate: 467
## Cell counts per day:
## Day08: 469 | Day14: 329 | Day20: 784
## [1] 1582  467
## [1] 467   3
## Iteration: 5 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 5 cells_candidate: 1585 muts_candidate: 463
## Cell counts per day:
## Day08: 482 | Day14: 362 | Day20: 741
## [1] 1585  463
## [1] 463   3
## Iteration: 6 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 6 cells_candidate: 1608 muts_candidate: 468
## Cell counts per day:
## Day08: 530 | Day14: 353 | Day20: 725
## [1] 1608  468
## [1] 468   3
## Iteration: 7 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 7 cells_candidate: 1545 muts_candidate: 463
## Cell counts per day:
## Day08: 473 | Day14: 346 | Day20: 726
## [1] 1545  463
## [1] 463   3
## Iteration: 8 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 8 cells_candidate: 1622 muts_candidate: 467
## Cell counts per day:
## Day08: 479 | Day14: 364 | Day20: 779
## [1] 1622  467
## [1] 467   3
## Iteration: 9 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 9 cells_candidate: 1628 muts_candidate: 463
## Cell counts per day:
## Day08: 528 | Day14: 356 | Day20: 744
## [1] 1628  463
## [1] 463   3
## Iteration: 10 loss_cell_indices: 704 loss_mut_indices: 239
```

```
## Iteration: 10 cells_candidate: 1560 muts_candidate: 464
## Cell counts per day:
## Day08: 484 | Day14: 355 | Day20: 721
## [1] 1560  464
## [1] 464   3
## Iteration: 11 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 11 cells_candidate: 1556 muts_candidate: 467
## Cell counts per day:
## Day08: 470 | Day14: 362 | Day20: 724
## [1] 1556  467
## [1] 467   3
## Iteration: 12 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 12 cells_candidate: 1660 muts_candidate: 474
## Cell counts per day:
## Day08: 540 | Day14: 342 | Day20: 778
## [1] 1660  474
## [1] 474   3
## Iteration: 13 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 13 cells_candidate: 1550 muts_candidate: 470
## Cell counts per day:
## Day08: 473 | Day14: 361 | Day20: 716
## [1] 1550  470
## [1] 470   3
## Iteration: 14 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 14 cells_candidate: 1576 muts_candidate: 462
## Cell counts per day:
## Day08: 474 | Day14: 352 | Day20: 750
## [1] 1576  462
## [1] 462   3
## Iteration: 15 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 15 cells_candidate: 1622 muts_candidate: 468
## Cell counts per day:
## Day08: 544 | Day14: 356 | Day20: 722
## [1] 1622  468
## [1] 468   3
## Iteration: 16 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 16 cells_candidate: 1599 muts_candidate: 467
## Cell counts per day:
## Day08: 481 | Day14: 346 | Day20: 772
## [1] 1599  467
## [1] 467   3
## Iteration: 17 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 17 cells_candidate: 1553 muts_candidate: 465
## Cell counts per day:
## Day08: 490 | Day14: 350 | Day20: 713
## [1] 1553  465
## [1] 465   3
## Iteration: 18 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 18 cells_candidate: 1617 muts_candidate: 466
## Cell counts per day:
## Day08: 540 | Day14: 365 | Day20: 712
## [1] 1617  466
## [1] 466   3
```

```
## Iteration: 19 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 19 cells_candidate: 1569 muts_candidate: 463
## Cell counts per day:
## Day08: 483 | Day14: 352 | Day20: 734
## [1] 1569  463
## [1] 463   3
## Iteration: 20 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 20 cells_candidate: 1606 muts_candidate: 469
## Cell counts per day:
## Day08: 475 | Day14: 356 | Day20: 775
## [1] 1606  469
## [1] 469   3
## Iteration: 21 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 21 cells_candidate: 1615 muts_candidate: 470
## Cell counts per day:
## Day08: 536 | Day14: 353 | Day20: 726
## [1] 1615  470
## [1] 470   3
## Iteration: 22 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 22 cells_candidate: 1568 muts_candidate: 458
## Cell counts per day:
## Day08: 484 | Day14: 372 | Day20: 712
## [1] 1568  458
## [1] 458   3
## Iteration: 23 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 23 cells_candidate: 1560 muts_candidate: 460
## Cell counts per day:
## Day08: 475 | Day14: 374 | Day20: 711
## [1] 1560  460
## [1] 460   3
## Iteration: 24 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 24 cells_candidate: 1650 muts_candidate: 467
## Cell counts per day:
## Day08: 532 | Day14: 351 | Day20: 767
## [1] 1650  467
## [1] 467   3
## Iteration: 25 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 25 cells_candidate: 1590 muts_candidate: 466
## Cell counts per day:
## Day08: 489 | Day14: 366 | Day20: 735
## [1] 1590  466
## [1] 466   3
## Iteration: 26 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 26 cells_candidate: 1577 muts_candidate: 467
## Cell counts per day:
## Day08: 493 | Day14: 360 | Day20: 724
## [1] 1577  467
## [1] 467   3
## Iteration: 27 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 27 cells_candidate: 1609 muts_candidate: 466
## Cell counts per day:
## Day08: 537 | Day14: 343 | Day20: 729
## [1] 1609  466
```

```
## [1] 466   3
## Iteration: 28 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 28 cells_candidate: 1636 muts_candidate: 464
## Cell counts per day:
## Day08: 486 | Day14: 376 | Day20: 774
## [1] 1636  464
## [1] 464   3
## Iteration: 29 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 29 cells_candidate: 1579 muts_candidate: 456
## Cell counts per day:
## Day08: 478 | Day14: 362 | Day20: 739
## [1] 1579  456
## [1] 456   3
## Iteration: 30 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 30 cells_candidate: 1619 muts_candidate: 467
## Cell counts per day:
## Day08: 540 | Day14: 349 | Day20: 730
## [1] 1619  467
## [1] 467   3
## Iteration: 31 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 31 cells_candidate: 1542 muts_candidate: 468
## Cell counts per day:
## Day08: 465 | Day14: 349 | Day20: 728
## [1] 1542  468
## [1] 468   3
## Iteration: 32 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 32 cells_candidate: 1620 muts_candidate: 471
## Cell counts per day:
## Day08: 494 | Day14: 371 | Day20: 755
## [1] 1620  471
## [1] 471   3
## Iteration: 33 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 33 cells_candidate: 1627 muts_candidate: 468
## Cell counts per day:
## Day08: 535 | Day14: 369 | Day20: 723
## [1] 1627  468
## [1] 468   3
## Iteration: 34 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 34 cells_candidate: 1589 muts_candidate: 467
## Cell counts per day:
## Day08: 487 | Day14: 360 | Day20: 742
## [1] 1589  467
## [1] 467   3
## Iteration: 35 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 35 cells_candidate: 1561 muts_candidate: 459
## Cell counts per day:
## Day08: 476 | Day14: 351 | Day20: 734
## [1] 1561  459
## [1] 459   3
## Iteration: 36 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 36 cells_candidate: 1681 muts_candidate: 471
## Cell counts per day:
## Day08: 537 | Day14: 369 | Day20: 775
```

```
## [1] 1681  471
## [1] 471   3
## Iteration: 37 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 37 cells_candidate: 1577 muts_candidate: 460
## Cell counts per day:
## Day08: 484 | Day14: 358 | Day20: 735
## [1] 1577  460
## [1] 460   3
## Iteration: 38 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 38 cells_candidate: 1598 muts_candidate: 463
## Cell counts per day:
## Day08: 483 | Day14: 375 | Day20: 740
## [1] 1598  463
## [1] 463   3
## Iteration: 39 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 39 cells_candidate: 1649 muts_candidate: 469
## Cell counts per day:
## Day08: 550 | Day14: 367 | Day20: 732
## [1] 1649  469
## [1] 469   3
## Iteration: 40 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 40 cells_candidate: 1621 muts_candidate: 469
## Cell counts per day:
## Day08: 498 | Day14: 355 | Day20: 768
## [1] 1621  469
## [1] 469   3
## Iteration: 41 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 41 cells_candidate: 1524 muts_candidate: 465
## Cell counts per day:
## Day08: 475 | Day14: 333 | Day20: 716
## [1] 1524  465
## [1] 465   3
## Iteration: 42 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 42 cells_candidate: 1632 muts_candidate: 467
## Cell counts per day:
## Day08: 530 | Day14: 368 | Day20: 734
## [1] 1632  467
## [1] 467   3
## Iteration: 43 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 43 cells_candidate: 1588 muts_candidate: 463
## Cell counts per day:
## Day08: 485 | Day14: 370 | Day20: 733
## [1] 1588  463
## [1] 463   3
## Iteration: 44 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 44 cells_candidate: 1617 muts_candidate: 468
## Cell counts per day:
## Day08: 491 | Day14: 354 | Day20: 772
## [1] 1617  468
## [1] 468   3
## Iteration: 45 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 45 cells_candidate: 1603 muts_candidate: 465
## Cell counts per day:
```

```
## Day08: 524 | Day14: 353 | Day20: 726
## [1] 1603  465
## [1] 465   3
## Iteration: 46 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 46 cells_candidate: 1581 muts_candidate: 465
## Cell counts per day:
## Day08: 489 | Day14: 352 | Day20: 740
## [1] 1581  465
## [1] 465   3
## Iteration: 47 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 47 cells_candidate: 1550 muts_candidate: 461
## Cell counts per day:
## Day08: 482 | Day14: 355 | Day20: 713
## [1] 1550  461
## [1] 461   3
## Iteration: 48 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 48 cells_candidate: 1684 muts_candidate: 474
## Cell counts per day:
## Day08: 545 | Day14: 358 | Day20: 781
## [1] 1684  474
## [1] 474   3
## Iteration: 49 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 49 cells_candidate: 1583 muts_candidate: 468
## Cell counts per day:
## Day08: 478 | Day14: 360 | Day20: 745
## [1] 1583  468
## [1] 468   3
## Iteration: 50 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 50 cells_candidate: 1565 muts_candidate: 457
## Cell counts per day:
## Day08: 493 | Day14: 348 | Day20: 724
## [1] 1565  457
## [1] 457   3
## Iteration: 51 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 51 cells_candidate: 1641 muts_candidate: 469
## Cell counts per day:
## Day08: 544 | Day14: 376 | Day20: 721
## [1] 1641  469
## [1] 469   3
## Iteration: 52 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 52 cells_candidate: 1596 muts_candidate: 469
## Cell counts per day:
## Day08: 462 | Day14: 365 | Day20: 769
## [1] 1596  469
## [1] 469   3
## Iteration: 53 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 53 cells_candidate: 1600 muts_candidate: 468
## Cell counts per day:
## Day08: 486 | Day14: 372 | Day20: 742
## [1] 1600  468
## [1] 468   3
## Iteration: 54 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 54 cells_candidate: 1628 muts_candidate: 465
```

```
## Cell counts per day:
## Day08: 538 | Day14: 360 | Day20: 730
## [1] 1628  465
## [1] 465   3
## Iteration: 55 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 55 cells_candidate: 1581 muts_candidate: 468
## Cell counts per day:
## Day08: 491 | Day14: 361 | Day20: 729
## [1] 1581  468
## [1] 468   3
## Iteration: 56 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 56 cells_candidate: 1611 muts_candidate: 466
## Cell counts per day:
## Day08: 476 | Day14: 365 | Day20: 770
## [1] 1611  466
## [1] 466   3
## Iteration: 57 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 57 cells_candidate: 1614 muts_candidate: 467
## Cell counts per day:
## Day08: 537 | Day14: 344 | Day20: 733
## [1] 1614  467
## [1] 467   3
## Iteration: 58 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 58 cells_candidate: 1578 muts_candidate: 469
## Cell counts per day:
## Day08: 495 | Day14: 353 | Day20: 730
## [1] 1578  469
## [1] 469   3
## Iteration: 59 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 59 cells_candidate: 1581 muts_candidate: 466
## Cell counts per day:
## Day08: 487 | Day14: 359 | Day20: 735
## [1] 1581  466
## [1] 466   3
## Iteration: 60 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 60 cells_candidate: 1678 muts_candidate: 468
## Cell counts per day:
## Day08: 538 | Day14: 364 | Day20: 776
## [1] 1678  468
## [1] 468   3
## Iteration: 61 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 61 cells_candidate: 1549 muts_candidate: 464
## Cell counts per day:
## Day08: 481 | Day14: 344 | Day20: 724
## [1] 1549  464
## [1] 464   3
## Iteration: 62 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 62 cells_candidate: 1565 muts_candidate: 458
## Cell counts per day:
## Day08: 479 | Day14: 357 | Day20: 729
## [1] 1565  458
## [1] 458   3
## Iteration: 63 loss_cell_indices: 596 loss_mut_indices: 239
```

```
## Iteration: 63 cells_candidate: 1630 muts_candidate: 465
## Cell counts per day:
## Day08: 542 | Day14: 373 | Day20: 715
## [1] 1630  465
## [1] 465   3
## Iteration: 64 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 64 cells_candidate: 1640 muts_candidate: 465
## Cell counts per day:
## Day08: 492 | Day14: 375 | Day20: 773
## [1] 1640  465
## [1] 465   3
## Iteration: 65 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 65 cells_candidate: 1570 muts_candidate: 462
## Cell counts per day:
## Day08: 481 | Day14: 373 | Day20: 716
## [1] 1570  462
## [1] 462   3
## Iteration: 66 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 66 cells_candidate: 1619 muts_candidate: 464
## Cell counts per day:
## Day08: 540 | Day14: 356 | Day20: 723
## [1] 1619  464
## [1] 464   3
## Iteration: 67 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 67 cells_candidate: 1551 muts_candidate: 462
## Cell counts per day:
## Day08: 470 | Day14: 353 | Day20: 728
## [1] 1551  462
## [1] 462   3
## Iteration: 68 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 68 cells_candidate: 1615 muts_candidate: 467
## Cell counts per day:
## Day08: 485 | Day14: 351 | Day20: 779
## [1] 1615  467
## [1] 467   3
## Iteration: 69 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 69 cells_candidate: 1623 muts_candidate: 468
## Cell counts per day:
## Day08: 529 | Day14: 374 | Day20: 720
## [1] 1623  468
## [1] 468   3
## Iteration: 70 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 70 cells_candidate: 1558 muts_candidate: 461
## Cell counts per day:
## Day08: 484 | Day14: 363 | Day20: 711
## [1] 1558  461
## [1] 461   3
## Iteration: 71 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 71 cells_candidate: 1558 muts_candidate: 464
## Cell counts per day:
## Day08: 494 | Day14: 354 | Day20: 710
## [1] 1558  464
## [1] 464   3
```

```
## Iteration: 72 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 72 cells_candidate: 1657 muts_candidate: 475
## Cell counts per day:
## Day08: 534 | Day14: 364 | Day20: 759
## [1] 1657  475
## [1] 475   3
## Iteration: 73 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 73 cells_candidate: 1577 muts_candidate: 464
## Cell counts per day:
## Day08: 483 | Day14: 368 | Day20: 726
## [1] 1577  464
## [1] 464   3
## Iteration: 74 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 74 cells_candidate: 1580 muts_candidate: 459
## Cell counts per day:
## Day08: 495 | Day14: 379 | Day20: 706
## [1] 1580  459
## [1] 459   3
## Iteration: 75 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 75 cells_candidate: 1650 muts_candidate: 469
## Cell counts per day:
## Day08: 542 | Day14: 374 | Day20: 734
## [1] 1650  469
## [1] 469   3
## Iteration: 76 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 76 cells_candidate: 1594 muts_candidate: 469
## Cell counts per day:
## Day08: 469 | Day14: 349 | Day20: 776
## [1] 1594  469
## [1] 469   3
## Iteration: 77 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 77 cells_candidate: 1565 muts_candidate: 465
## Cell counts per day:
## Day08: 485 | Day14: 356 | Day20: 724
## [1] 1565  465
## [1] 465   3
## Iteration: 78 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 78 cells_candidate: 1602 muts_candidate: 464
## Cell counts per day:
## Day08: 540 | Day14: 363 | Day20: 699
## [1] 1602  464
## [1] 464   3
## Iteration: 79 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 79 cells_candidate: 1593 muts_candidate: 463
## Cell counts per day:
## Day08: 485 | Day14: 369 | Day20: 739
## [1] 1593  463
## [1] 463   3
## Iteration: 80 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 80 cells_candidate: 1631 muts_candidate: 466
## Cell counts per day:
## Day08: 497 | Day14: 370 | Day20: 764
## [1] 1631  466
```

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## [1] 466   3
## Iteration: 81 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 81 cells_candidate: 1610 muts_candidate: 471
## Cell counts per day:
## Day08: 536 | Day14: 340 | Day20: 734
## [1] 1610  471
## [1] 471   3
## Iteration: 82 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 82 cells_candidate: 1578 muts_candidate: 461
## Cell counts per day:
## Day08: 483 | Day14: 376 | Day20: 719
## [1] 1578  461
## [1] 461   3
## Iteration: 83 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 83 cells_candidate: 1592 muts_candidate: 465
## Cell counts per day:
## Day08: 496 | Day14: 374 | Day20: 722
## [1] 1592  465
## [1] 465   3
## Iteration: 84 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 84 cells_candidate: 1675 muts_candidate: 470
## Cell counts per day:
## Day08: 535 | Day14: 366 | Day20: 774
## [1] 1675  470
## [1] 470   3
## Iteration: 85 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 85 cells_candidate: 1584 muts_candidate: 469
## Cell counts per day:
## Day08: 488 | Day14: 361 | Day20: 735
## [1] 1584  469
## [1] 469   3
## Iteration: 86 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 86 cells_candidate: 1554 muts_candidate: 464
## Cell counts per day:
## Day08: 489 | Day14: 348 | Day20: 717
## [1] 1554  464
## [1] 464   3
## Iteration: 87 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 87 cells_candidate: 1641 muts_candidate: 466
## Cell counts per day:
## Day08: 536 | Day14: 373 | Day20: 732
## [1] 1641  466
## [1] 466   3
## Iteration: 88 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 88 cells_candidate: 1609 muts_candidate: 471
## Cell counts per day:
## Day08: 482 | Day14: 352 | Day20: 775
## [1] 1609  471
## [1] 471   3
## Iteration: 89 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 89 cells_candidate: 1566 muts_candidate: 464
## Cell counts per day:
## Day08: 497 | Day14: 360 | Day20: 709
```

```
## [1] 1566  464
## [1] 464   3
## Iteration: 90 loss_cell_indices: 595 loss_mut_indices: 239
## Iteration: 90 cells_candidate: 1643 muts_candidate: 462
## Cell counts per day:
## Day08: 540 | Day14: 370 | Day20: 733
## [1] 1643  462
## [1] 462   3
## Iteration: 91 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 91 cells_candidate: 1578 muts_candidate: 469
## Cell counts per day:
## Day08: 491 | Day14: 354 | Day20: 733
## [1] 1578  469
## [1] 469   3
## Iteration: 92 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 92 cells_candidate: 1630 muts_candidate: 471
## Cell counts per day:
## Day08: 493 | Day14: 363 | Day20: 774
## [1] 1630  471
## [1] 471   3
## Iteration: 93 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 93 cells_candidate: 1597 muts_candidate: 468
## Cell counts per day:
## Day08: 537 | Day14: 360 | Day20: 700
## [1] 1597  468
## [1] 468   3
## Iteration: 94 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 94 cells_candidate: 1576 muts_candidate: 467
## Cell counts per day:
## Day08: 480 | Day14: 365 | Day20: 731
## [1] 1576  467
## [1] 467   3
## Iteration: 95 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 95 cells_candidate: 1566 muts_candidate: 463
## Cell counts per day:
## Day08: 472 | Day14: 363 | Day20: 731
## [1] 1566  463
## [1] 463   3
## Iteration: 96 loss_cell_indices: 483 loss_mut_indices: 239
## Iteration: 96 cells_candidate: 1677 muts_candidate: 472
## Cell counts per day:
## Day08: 533 | Day14: 364 | Day20: 780
## [1] 1677  472
## [1] 472   3
## Iteration: 97 loss_cell_indices: 705 loss_mut_indices: 239
## Iteration: 97 cells_candidate: 1549 muts_candidate: 466
## Cell counts per day:
## Day08: 475 | Day14: 354 | Day20: 720
## [1] 1549  466
## [1] 466   3
## Iteration: 98 loss_cell_indices: 704 loss_mut_indices: 239
## Iteration: 98 cells_candidate: 1567 muts_candidate: 466
## Cell counts per day:
```

```
## Day08: 468 | Day14: 373 | Day20: 726
## [1] 1567  466
## [1] 466   3
## Iteration: 99 loss_cell_indices: 596 loss_mut_indices: 239
## Iteration: 99 cells_candidate: 1613 muts_candidate: 470
## Cell counts per day:
## Day08: 549 | Day14: 350 | Day20: 714
## [1] 1613  470
## [1] 470   3
## Iteration: 100 loss_cell_indices: 592 loss_mut_indices: 239
## Iteration: 100 cells_candidate: 1627 muts_candidate: 466
## Cell counts per day:
## Day08: 474 | Day14: 365 | Day20: 788
## [1] 1627  466
## [1] 466   3

# same cell number for day scMutrace est
# at least 100 cells
cell_day_map <- setNames(rowname_day$day, rowname_day$rowname)

for (iter in seq_len(iteration_times)) {

  # 1. sampling
  sampled_cells <- rowname_day %>%
    group_by(day) %>%
    slice_sample(n = 100, replace = TRUE) %>%
    ungroup() %>%
    pull(rowname)
  C_partial_missing <- data_check_CM[match(sampled_cells, rownames(data_check_CM)),, drop = FALSE]
  # 2. count
  MutPerCell <- rowSums(C_partial_missing == "Mut")
  CellPerMut <- colSums(C_partial_missing == "Mut")
  cells_candidate <- rownames(C_partial_missing)[MutPerCell >= cell_num]
  muts_candidate <- colnames(C_partial_missing)[CellPerMut >= mut_num]
  data_cell_mut_raw_check <- C_partial_missing[cells_candidate, muts_candidate, drop = FALSE]

  # logging
  tab <- table(sub(".*_(Day[0-9]+)_.*", "\\1", rownames(data_cell_mut_raw_check)))
  message(paste0("Iteration: ", iter,
                 " sampled_cells: ", length(sampled_cells),
                 " cells_candidate: ", length(cells_candidate),
                 " muts_candidate: ", length(muts_candidate)))
  message(paste(names(tab), tab, sep = ": ", collapse = " | "))

  # 3. Mut × Time
  M_partial_missing <- matrix(0,
                              nrow = length(muts_candidate),
                              ncol = length(time_types),
                              dimnames = list(muts_candidate, time_types))

  # cal
  for (mut in muts_candidate) {
    cells_with_mut <- rownames(data_cell_mut_raw_check)[data_cell_mut_raw_check[, mut] == "Mut"]
```

```
    cells_with_mut_clean <- sub("\\.[0-9]+$", "", cells_with_mut)
    days <- cell_day_map[cells_with_mut_clean]
    M_partial_missing[mut, ] <- table(factor(days, levels = time_types))
  }

  # 4. scMutScore
  res_index_tmp <- cal_scMutScore(data_cell_mut_raw_check, M_partial_missing)$score
  res_index_tmp$iteration <- paste0("iter", iter)
  res_index_tmp <- res_index_tmp[res_index_tmp$Diversity > 0, ]
  res_index_tmp$Cellbarcode <- rownames(res_index_tmp)
  rownames(res_index_tmp) <- NULL
  data_scMutScore_list[[iter]] <- res_index_tmp
}
## Iteration: 1 sampled_cells: 300 cells_candidate: 300 muts_candidate: 224
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 224
## [1] 224   3
## Iteration: 2 sampled_cells: 300 cells_candidate: 300 muts_candidate: 220
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 220
## [1] 220   3
## Iteration: 3 sampled_cells: 300 cells_candidate: 300 muts_candidate: 231
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 231
## [1] 231   3
## Iteration: 4 sampled_cells: 300 cells_candidate: 300 muts_candidate: 216
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 216
## [1] 216   3
## Iteration: 5 sampled_cells: 300 cells_candidate: 300 muts_candidate: 226
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 226
## [1] 226   3
## Iteration: 6 sampled_cells: 300 cells_candidate: 300 muts_candidate: 223
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 223
## [1] 223   3
## Iteration: 7 sampled_cells: 300 cells_candidate: 300 muts_candidate: 241
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 241
## [1] 241   3
## Iteration: 8 sampled_cells: 300 cells_candidate: 300 muts_candidate: 220
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 220
## [1] 220   3
## Iteration: 9 sampled_cells: 300 cells_candidate: 300 muts_candidate: 227
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 227
## [1] 227   3
## Iteration: 10 sampled_cells: 300 cells_candidate: 300 muts_candidate: 231
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 231
## [1] 231   3
```

```
## Iteration: 11 sampled_cells: 300 cells_candidate: 300 muts_candidate: 225
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 225
## [1] 225   3
## Iteration: 12 sampled_cells: 300 cells_candidate: 300 muts_candidate: 214
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 214
## [1] 214   3
## Iteration: 13 sampled_cells: 300 cells_candidate: 300 muts_candidate: 225
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 225
## [1] 225   3
## Iteration: 14 sampled_cells: 300 cells_candidate: 300 muts_candidate: 233
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 233
## [1] 233   3
## Iteration: 15 sampled_cells: 300 cells_candidate: 300 muts_candidate: 238
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 238
## [1] 238   3
## Iteration: 16 sampled_cells: 300 cells_candidate: 300 muts_candidate: 219
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 219
## [1] 219   3
## Iteration: 17 sampled_cells: 300 cells_candidate: 300 muts_candidate: 229
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 229
## [1] 229   3
## Iteration: 18 sampled_cells: 300 cells_candidate: 300 muts_candidate: 229
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 229
## [1] 229   3
## Iteration: 19 sampled_cells: 300 cells_candidate: 300 muts_candidate: 232
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 232
## [1] 232   3
## Iteration: 20 sampled_cells: 300 cells_candidate: 300 muts_candidate: 216
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 216
## [1] 216   3
## Iteration: 21 sampled_cells: 300 cells_candidate: 300 muts_candidate: 221
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 221
## [1] 221   3
## Iteration: 22 sampled_cells: 300 cells_candidate: 300 muts_candidate: 245
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 245
## [1] 245   3
## Iteration: 23 sampled_cells: 300 cells_candidate: 300 muts_candidate: 213
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 213
## [1] 213   3
## Iteration: 24 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
```

```
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 25 sampled_cells: 300 cells_candidate: 300 muts_candidate: 217
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 217
## [1] 217   3
## Iteration: 26 sampled_cells: 300 cells_candidate: 300 muts_candidate: 215
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 215
## [1] 215   3
## Iteration: 27 sampled_cells: 300 cells_candidate: 300 muts_candidate: 218
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 218
## [1] 218   3
## Iteration: 28 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 29 sampled_cells: 300 cells_candidate: 300 muts_candidate: 237
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 237
## [1] 237   3
## Iteration: 30 sampled_cells: 300 cells_candidate: 300 muts_candidate: 234
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 234
## [1] 234   3
## Iteration: 31 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 32 sampled_cells: 300 cells_candidate: 300 muts_candidate: 227
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 227
## [1] 227   3
## Iteration: 33 sampled_cells: 300 cells_candidate: 300 muts_candidate: 212
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 212
## [1] 212   3
## Iteration: 34 sampled_cells: 300 cells_candidate: 300 muts_candidate: 234
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 234
## [1] 234   3
## Iteration: 35 sampled_cells: 300 cells_candidate: 300 muts_candidate: 217
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 217
## [1] 217   3
## Iteration: 36 sampled_cells: 300 cells_candidate: 300 muts_candidate: 222
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 222
## [1] 222   3
## Iteration: 37 sampled_cells: 300 cells_candidate: 300 muts_candidate: 231
## Day08: 100 | Day14: 100 | Day20: 100
```

```
## [1] 300 231
## [1] 231   3
## Iteration: 38 sampled_cells: 300 cells_candidate: 300 muts_candidate: 227
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 227
## [1] 227   3
## Iteration: 39 sampled_cells: 300 cells_candidate: 300 muts_candidate: 227
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 227
## [1] 227   3
## Iteration: 40 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 41 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 42 sampled_cells: 300 cells_candidate: 300 muts_candidate: 230
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 230
## [1] 230   3
## Iteration: 43 sampled_cells: 300 cells_candidate: 300 muts_candidate: 221
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 221
## [1] 221   3
## Iteration: 44 sampled_cells: 300 cells_candidate: 300 muts_candidate: 211
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 211
## [1] 211   3
## Iteration: 45 sampled_cells: 300 cells_candidate: 300 muts_candidate: 216
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 216
## [1] 216   3
## Iteration: 46 sampled_cells: 300 cells_candidate: 300 muts_candidate: 224
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 224
## [1] 224   3
## Iteration: 47 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 48 sampled_cells: 300 cells_candidate: 300 muts_candidate: 231
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 231
## [1] 231   3
## Iteration: 49 sampled_cells: 300 cells_candidate: 300 muts_candidate: 233
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 233
## [1] 233   3
## Iteration: 50 sampled_cells: 300 cells_candidate: 300 muts_candidate: 219
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 219
```

```
## [1] 219   3
## Iteration: 51 sampled_cells: 300 cells_candidate: 300 muts_candidate: 221
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 221
## [1] 221   3
## Iteration: 52 sampled_cells: 300 cells_candidate: 300 muts_candidate: 221
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 221
## [1] 221   3
## Iteration: 53 sampled_cells: 300 cells_candidate: 300 muts_candidate: 222
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 222
## [1] 222   3
## Iteration: 54 sampled_cells: 300 cells_candidate: 300 muts_candidate: 218
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 218
## [1] 218   3
## Iteration: 55 sampled_cells: 300 cells_candidate: 300 muts_candidate: 216
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 216
## [1] 216   3
## Iteration: 56 sampled_cells: 300 cells_candidate: 300 muts_candidate: 232
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 232
## [1] 232   3
## Iteration: 57 sampled_cells: 300 cells_candidate: 300 muts_candidate: 216
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 216
## [1] 216   3
## Iteration: 58 sampled_cells: 300 cells_candidate: 300 muts_candidate: 226
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 226
## [1] 226   3
## Iteration: 59 sampled_cells: 300 cells_candidate: 300 muts_candidate: 206
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 206
## [1] 206   3
## Iteration: 60 sampled_cells: 300 cells_candidate: 300 muts_candidate: 219
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 219
## [1] 219   3
## Iteration: 61 sampled_cells: 300 cells_candidate: 300 muts_candidate: 235
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 235
## [1] 235   3
## Iteration: 62 sampled_cells: 300 cells_candidate: 300 muts_candidate: 235
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 235
## [1] 235   3
## Iteration: 63 sampled_cells: 300 cells_candidate: 300 muts_candidate: 234
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 234
## [1] 234   3
```

```
## Iteration: 64 sampled_cells: 300 cells_candidate: 300 muts_candidate: 231
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 231
## [1] 231   3
## Iteration: 65 sampled_cells: 300 cells_candidate: 300 muts_candidate: 216
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 216
## [1] 216   3
## Iteration: 66 sampled_cells: 300 cells_candidate: 300 muts_candidate: 220
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 220
## [1] 220   3
## Iteration: 67 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 68 sampled_cells: 300 cells_candidate: 300 muts_candidate: 220
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 220
## [1] 220   3
## Iteration: 69 sampled_cells: 300 cells_candidate: 300 muts_candidate: 223
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 223
## [1] 223   3
## Iteration: 70 sampled_cells: 300 cells_candidate: 300 muts_candidate: 209
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 209
## [1] 209   3
## Iteration: 71 sampled_cells: 300 cells_candidate: 300 muts_candidate: 229
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 229
## [1] 229   3
## Iteration: 72 sampled_cells: 300 cells_candidate: 300 muts_candidate: 213
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 213
## [1] 213   3
## Iteration: 73 sampled_cells: 300 cells_candidate: 300 muts_candidate: 215
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 215
## [1] 215   3
## Iteration: 74 sampled_cells: 300 cells_candidate: 300 muts_candidate: 229
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 229
## [1] 229   3
## Iteration: 75 sampled_cells: 300 cells_candidate: 300 muts_candidate: 220
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 220
## [1] 220   3
## Iteration: 76 sampled_cells: 300 cells_candidate: 300 muts_candidate: 225
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 225
## [1] 225   3
## Iteration: 77 sampled_cells: 300 cells_candidate: 300 muts_candidate: 225
```

```
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 225
## [1] 225   3
## Iteration: 78 sampled_cells: 300 cells_candidate: 300 muts_candidate: 217
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 217
## [1] 217   3
## Iteration: 79 sampled_cells: 300 cells_candidate: 300 muts_candidate: 223
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 223
## [1] 223   3
## Iteration: 80 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 228
## [1] 228   3
## Iteration: 81 sampled_cells: 300 cells_candidate: 300 muts_candidate: 227
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 227
## [1] 227   3
## Iteration: 82 sampled_cells: 300 cells_candidate: 300 muts_candidate: 237
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 237
## [1] 237   3
## Iteration: 83 sampled_cells: 300 cells_candidate: 300 muts_candidate: 217
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 217
## [1] 217   3
## Iteration: 84 sampled_cells: 300 cells_candidate: 300 muts_candidate: 219
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 219
## [1] 219   3
## Iteration: 85 sampled_cells: 300 cells_candidate: 300 muts_candidate: 221
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 221
## [1] 221   3
## Iteration: 86 sampled_cells: 300 cells_candidate: 300 muts_candidate: 223
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 223
## [1] 223   3
## Iteration: 87 sampled_cells: 300 cells_candidate: 300 muts_candidate: 236
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 236
## [1] 236   3
## Iteration: 88 sampled_cells: 300 cells_candidate: 300 muts_candidate: 237
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 237
## [1] 237   3
## Iteration: 89 sampled_cells: 300 cells_candidate: 300 muts_candidate: 232
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 232
## [1] 232   3
## Iteration: 90 sampled_cells: 300 cells_candidate: 300 muts_candidate: 228
## Day08: 100 | Day14: 100 | Day20: 100
```

```
## [1] 300 228
## [1] 228   3
## Iteration: 91 sampled_cells: 300 cells_candidate: 300 muts_candidate: 227
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 227
## [1] 227   3
## Iteration: 92 sampled_cells: 300 cells_candidate: 300 muts_candidate: 221
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 221
## [1] 221   3
## Iteration: 93 sampled_cells: 300 cells_candidate: 300 muts_candidate: 240
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 240
## [1] 240   3
## Iteration: 94 sampled_cells: 300 cells_candidate: 300 muts_candidate: 201
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 201
## [1] 201   3
## Iteration: 95 sampled_cells: 300 cells_candidate: 300 muts_candidate: 223
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 223
## [1] 223   3
## Iteration: 96 sampled_cells: 300 cells_candidate: 300 muts_candidate: 227
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 227
## [1] 227   3
## Iteration: 97 sampled_cells: 300 cells_candidate: 300 muts_candidate: 222
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 222
## [1] 222   3
## Iteration: 98 sampled_cells: 300 cells_candidate: 300 muts_candidate: 241
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 241
## [1] 241   3
## Iteration: 99 sampled_cells: 300 cells_candidate: 300 muts_candidate: 221
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 221
## [1] 221   3
## Iteration: 100 sampled_cells: 300 cells_candidate: 300 muts_candidate: 222
## Day08: 100 | Day14: 100 | Day20: 100
## [1] 300 222
## [1] 222   3

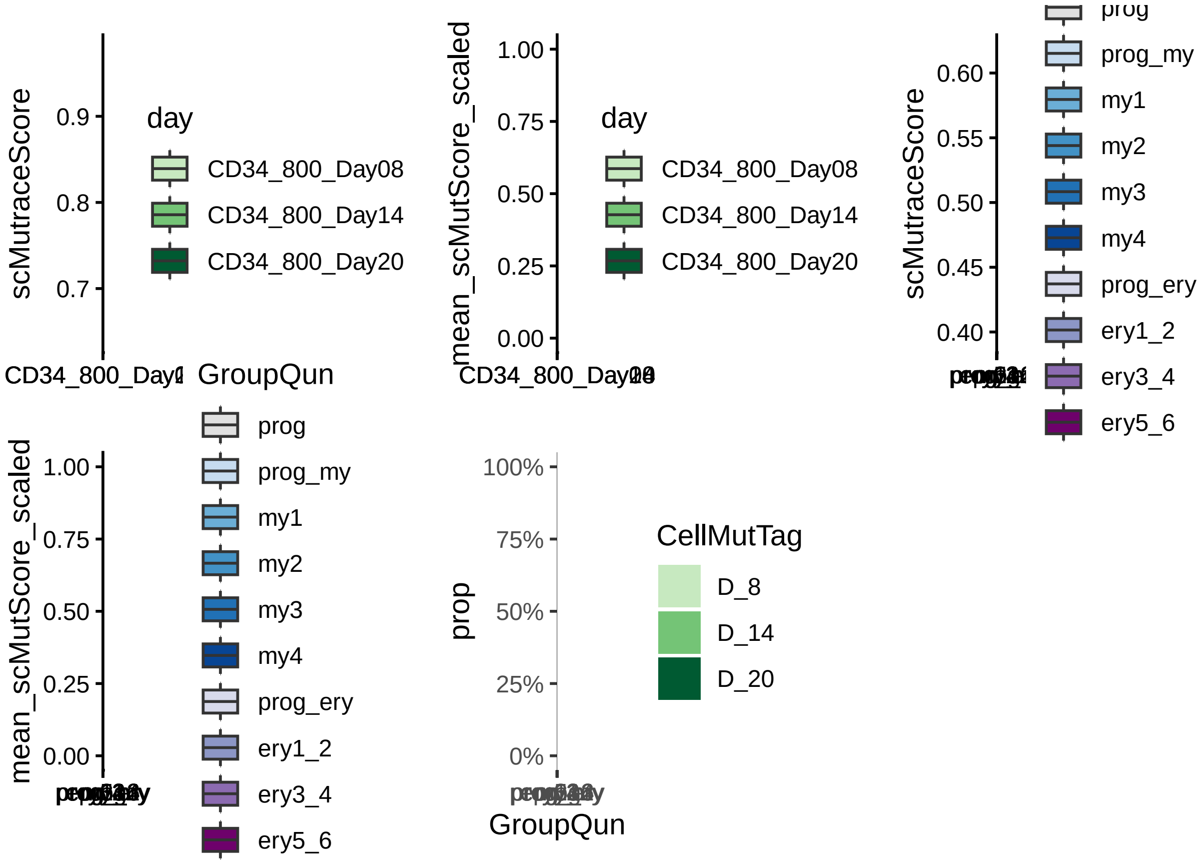
# 3. merge res
data_scMutModel_res  <- do.call(rbind, data_scMutModel_list)
data_scMutScore_res <- do.call(rbind, data_scMutScore_list)
data_scMutScore_res$day <- sub("^(CD34_800_Day[0-9]+)_.*", "\\1", data_scMutScore_res$Cellbarcode)

# 4. column
data_scMutModel_res <- data_scMutModel_res %>%
  left_join(data_check_Meta, by = "Cellbarcode")

data_scMutModel_res <- data_scMutModel_res %>% dplyr::filter(Group %in% c("Prog", "prog_my", "prog_ery"
```

```r
table(data_check_Meta$Group)
##
##     ery1     ery2     ery3     ery4     ery5     ery6      mix      my1
##       59       83       61       67       63       94       37      161
##      my2      my3      my4     prog prog_ery  prog_my
##      365      429      308      122       18       26
# keep 100 cells for each cell type

data_scMutModel_res$GroupQun <- as.character(data_scMutModel_res$Group)

data_scMutModel_res$GroupQun[grepl("ery1|ery2", data_scMutModel_res$Group)] <- "ery1_2"
data_scMutModel_res$GroupQun[grepl("ery3|ery4", data_scMutModel_res$Group)] <- "ery3_4"
data_scMutModel_res$GroupQun[grepl("ery5|ery6", data_scMutModel_res$Group)] <- "ery5_6"

data_scMutModel_res$Group <- factor(data_scMutModel_res$Group, levels = c("Prog", "prog", "prog_my", "my
data_scMutModel_res$GroupQun <- factor(data_scMutModel_res$GroupQun, levels = c("Prog", "prog", "prog_my
data_scMutModel_res$CellMutTag <- factor(data_scMutModel_res$CellMutTag, levels = c("D_8", "D_14", "D_20

# 5. plot
data_scMutModel_res_summary <- data_scMutModel_res %>%
  dplyr::group_by(iteration, GroupQun) %>%
  dplyr::summarise(mean_scMutScore = mean(scMutScore, na.rm = TRUE), mean_EpiTraceAge = mean(EpiTraceAge
  dplyr::ungroup()
## `summarise()` has grouped output by 'iteration'. You can override using the
## `.groups` argument.

data_scMutModel_res_summary <- data_scMutModel_res_summary %>%
  dplyr::mutate(
    mean_scMutScore_scaled = scales::rescale(mean_scMutScore, to = c(0, 1)),
    mean_EpiTraceAge_scaled = scales::rescale(mean_EpiTraceAge, to = c(0, 1))
  )

data_scMutModel_res_summary_Day <- data_scMutScore_res %>%
  dplyr::group_by(iteration, day) %>%
  dplyr::summarise(mean_scMutScore = mean(scMutScore, na.rm = TRUE)) %>%
  dplyr::ungroup()
## `summarise()` has grouped output by 'iteration'. You can override using the
## `.groups` argument.

data_scMutModel_res_summary_Day <- data_scMutModel_res_summary_Day %>%
  dplyr::mutate(
    mean_scMutScore_scaled = scales::rescale(mean_scMutScore, to = c(0, 1))
  )

data_stability <- data_scMutModel_res %>%
  dplyr::group_by(GroupQun) %>%
  dplyr::summarise(
    mean_score = mean(scMutScore, na.rm = TRUE),
    sd_score = sd(scMutScore, na.rm = TRUE),
    cv_score = sd_score / mean_score
  )

p_scMutrace_iter_GroupQun <- ggplot(data = data_scMutModel_res_summary, mapping = aes(x = GroupQun, y =
```

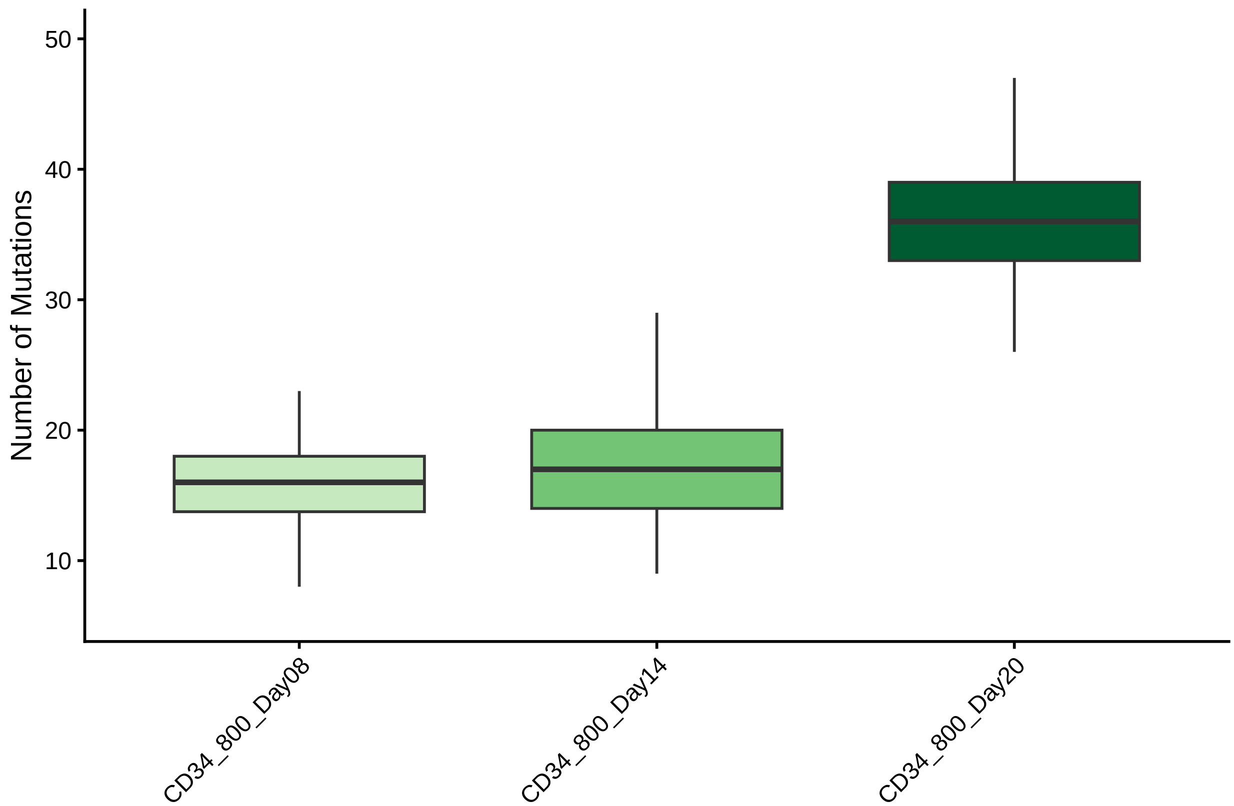
```
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()

p_scMutrace_iter_GroupQunScaled <- ggplot(data = data_scMutModel_res_summary, mapping = aes(x = GroupQu
  geom_boxplot(outliers = FALSE) +
  ylim(0, 1) +
  labs(x = "", y = "mean_scMutScore_scaled") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()

 p_scMutrace_iter_Day <- ggplot(data = data_scMutModel_res_summary_Day, mapping = aes(x = day, y = mean_
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = c(CD34_800_Day08 = "#c7e9c0", CD34_800_Day14 = "#74c476", CD34_800_Day20 =
  theme_classic()

 p_scMutrace_iter_Day_scaled <- ggplot(data = data_scMutModel_res_summary_Day, mapping = aes(x = day, y
  geom_boxplot(outliers = FALSE) +
  ylim(0, 1) +
  labs(x = "", y = "mean_scMutScore_scaled") +
  scale_fill_manual(values = c(CD34_800_Day08 = "#c7e9c0", CD34_800_Day14 = "#74c476", CD34_800_Day20 =
  theme_classic()

df_iter_frac <- data_scMutModel_res %>%
  group_by(GroupQun, CellMutTag) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(GroupQun) %>%
  mutate(prop = n / sum(n))

p_iter_cellfrac <- ggplot(df_iter_frac, aes(x = GroupQun, y = prop, fill = CellMutTag)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_fill_manual(values = day_color) +
  scale_y_continuous(labels = scales::percent) +
  theme_bw()

p_scMutrace_iter_Day + p_scMutrace_iter_Day_scaled + p_scMutrace_iter_GroupQun + p_scMutrace_iter_Group
```



10 randomMutNum

```
set.seed(2025)
# at least 100 cells
sample_size <- 100
iteration_times <- 100
iter_num <- 1:iteration_times
cell_num <- 1
mut_num <- 1

data_random_mutNum <- data.frame(iteration_num = iter_num, CD34_800_Day08 = 0,
                                 CD34_800_Day14 = 0, CD34_800_Day20 = 0)

CD34_Meta_selectedRandom <- CD34_Meta[CD34_Meta$Cellbarcode %in% rownames(CD34_CSM),]

for (iter in iter_num){
  cell_08 <- CD34_Meta_selectedRandom %>% filter(CellMutTag == "D_8") %>% pull(Cellbarcode) %>% sample(
  cell_14 <- CD34_Meta_selectedRandom %>% filter(CellMutTag == "D_14") %>% pull(Cellbarcode) %>% sample
  cell_20 <- CD34_Meta_selectedRandom %>% filter(CellMutTag == "D_20") %>% pull(Cellbarcode) %>% sample
  mut_08 <- retriveLocsByCells(CD34_CSM, cell_08)$Mut
  mut_14 <- retriveLocsByCells(CD34_CSM, cell_14)$Mut
  mut_20 <- retriveLocsByCells(CD34_CSM, cell_20)$Mut
  data_random_mutNum[iter, "CD34_800_Day08"] <- length(mut_08)
```

```
  data_random_mutNum[iter, "CD34_800_Day14"] <- length(mut_14)
  data_random_mutNum[iter, "CD34_800_Day20"] <- length(mut_20)
}

data_random_mutNum_long <- data_random_mutNum %>%
  pivot_longer(
    cols = -iteration_num,
    names_to = "Sample",
    values_to = "Mutation_Number"
  )

p_randomMutNum_est <- ggplot(data_random_mutNum_long, aes(x = Sample, y = Mutation_Number, fill = Sampl
  geom_boxplot(outlier.shape = NA, width = 0.7) +
  scale_fill_manual(values = c(CD34_800_Day08 = "#c7e9c0", CD34_800_Day14 = "#74c476", CD34_800_Day20 =
  #geom_jitter(width = 0.15, size = 1.5, alpha = 0.6) +
  theme_classic() +
  labs(
    x = "",
    y = "Number of Mutations"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )

p_randomMutNum_est
```



11 outputFigures

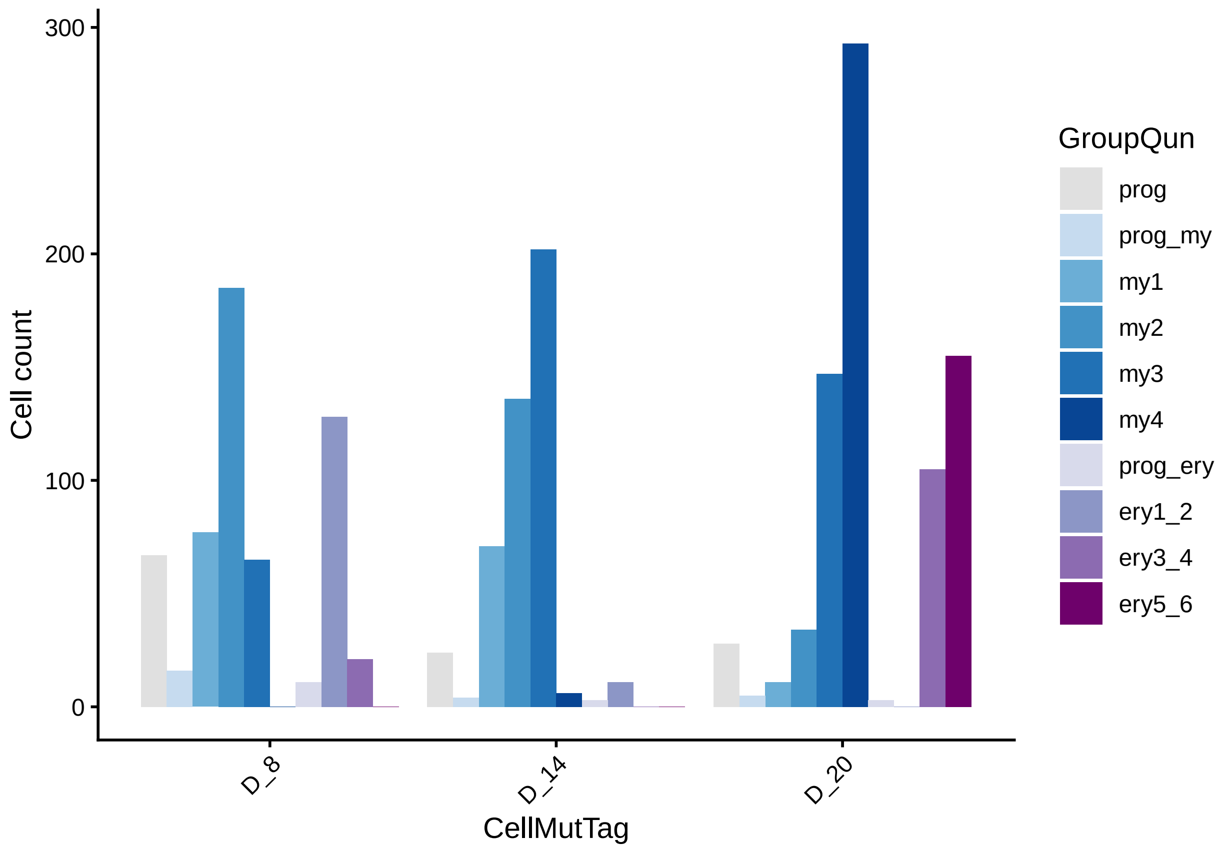
```
res_index <- res_index[res_index$Diversity > 0, ]

res_index %>% dplyr::filter(Group %in% c("Prog", "prog_my", "prog_ery", "prog", "my1","my2",'my3','my4')
## `summarise()` has grouped output by 'clone', 'Group'. You can override using
## the `.groups` argument.
res_index_meta_summary_scMutScore$CultureDay <- paste("D_", res_index_meta_summary_scMutScore$cell_day,

res_index_meta_summary_scMutScore$cell_group <- 'Prog'
res_index_meta_summary_scMutScore$cell_group[grepl('my1',res_index_meta_summary_scMutScore$Group)] <- 'I
res_index_meta_summary_scMutScore$cell_group[grepl('my2|my3|my4',res_index_meta_summary_scMutScore$Grou
res_index_meta_summary_scMutScore$cell_group <- factor(res_index_meta_summary_scMutScore$cell_group,leve

res_index$GroupQun <- as.character(res_index$Group)

res_index$GroupQun[grepl("ery1|ery2", res_index$Group)] <- "ery1_2"
res_index$GroupQun[grepl("ery3|ery4", res_index$Group)] <- "ery3_4"
res_index$GroupQun[grepl("ery5|ery6", res_index$Group)] <- "ery5_6"

res_index$Group <- factor(res_index$Group, levels = c("Prog", "prog", "prog_my", "my1", "my2", "my3", "r
res_index$GroupQun <- factor(res_index$GroupQun, levels = c("Prog", "prog", "prog_my", "my1", "my2", "my
res_index_meta_summary_scMutScore$CultureDay <- factor(res_index_meta_summary_scMutScore$CultureDay, le
```

```r
CD34_Meta_selected <- CD34_Meta[CD34_Meta$Cellbarcode %in% res_index$Cellbarcode,]
CD34_Meta_selected$GroupQun <- CD34_Meta_selected$Group
CD34_Meta_selected$GroupQun[grepl("ery1|ery2", CD34_Meta_selected$Group)] <- "ery1_2"
CD34_Meta_selected$GroupQun[grepl("ery3|ery4", CD34_Meta_selected$Group)] <- "ery3_4"
CD34_Meta_selected$GroupQun[grepl("ery5|ery6", CD34_Meta_selected$Group)] <- "ery5_6"

UMAP_Day8 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 8),
             aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

UMAP_Day14 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 14),
             aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

UMAP_Day20 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 20),
             aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_Days <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = CellMutTag)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = day_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_Groups <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = Group)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_GroupsQun <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = GroupQun)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_pseudutime_my <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color =  pseuduti
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option = "plasma") +
  xlab("") + ylab("") +
```

```
  theme_classic()

p_UMAP_pseudutime_ery <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color =  pseudut
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option = "plasma") +
  xlab("") + ylab("") +
  theme_classic()

# plot scMutScore by days
p_scMutScorebydays <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=scMutScore, fill = CellMutTag)) +
  scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

# plot scMutScore by cell types
p_scMutScorebycelltypes <- ggplot(res_index, aes(x=as.factor(Group), y=scMutScore, fill = Group)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_scMutScorebycelltypesQun <- ggplot(res_index, aes(x=as.factor(GroupQun), y=scMutScore, fill = GroupQu
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  ylim(0,1) +
  theme_classic()

p_EpiScorebyCelltype <- ggplot(res_index, aes(x=as.factor(Group), y=EpiTraceAge_iterative, fill = Group
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_EpiScorebyCelltypeQun <- ggplot(res_index, aes(x=as.factor(GroupQun), y=EpiTraceAge_iterative, fill =
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_CloneLevelByCellGroup <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = as.factor(cell_g
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone),  alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

p_CloneLevelByDaysSplitbyCellgroup <- ggplot(res_index_meta_summary_scMutScore,aes(x=factor(cell_day),y
  geom_line(aes(group=clone),size=0.5,alpha=0.1,linetype='dashed') + geom_boxplot(outlier.alpha = 1,fil
  facet_wrap(~cell_group)  + theme_classic() + xlab('Days in culture') + ylab('Mean scMutScore') + them
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

p_CloneLevelByDaysSplitbyCellType <-  ggplot(res_index_meta_summary_scMutScore,aes(x=factor(cell_day),y
  geom_line(aes(group=clone),size=0.5,alpha=0.1,linetype='dashed')  + geom_boxplot(outlier.alpha = 1,fi
  facet_wrap(~Group)  + theme_classic() + xlab('Days in culture') + ylab('Mean scMutScore') + theme(tex

p_CloneLevelByDays <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = as.factor(CultureDay)
```

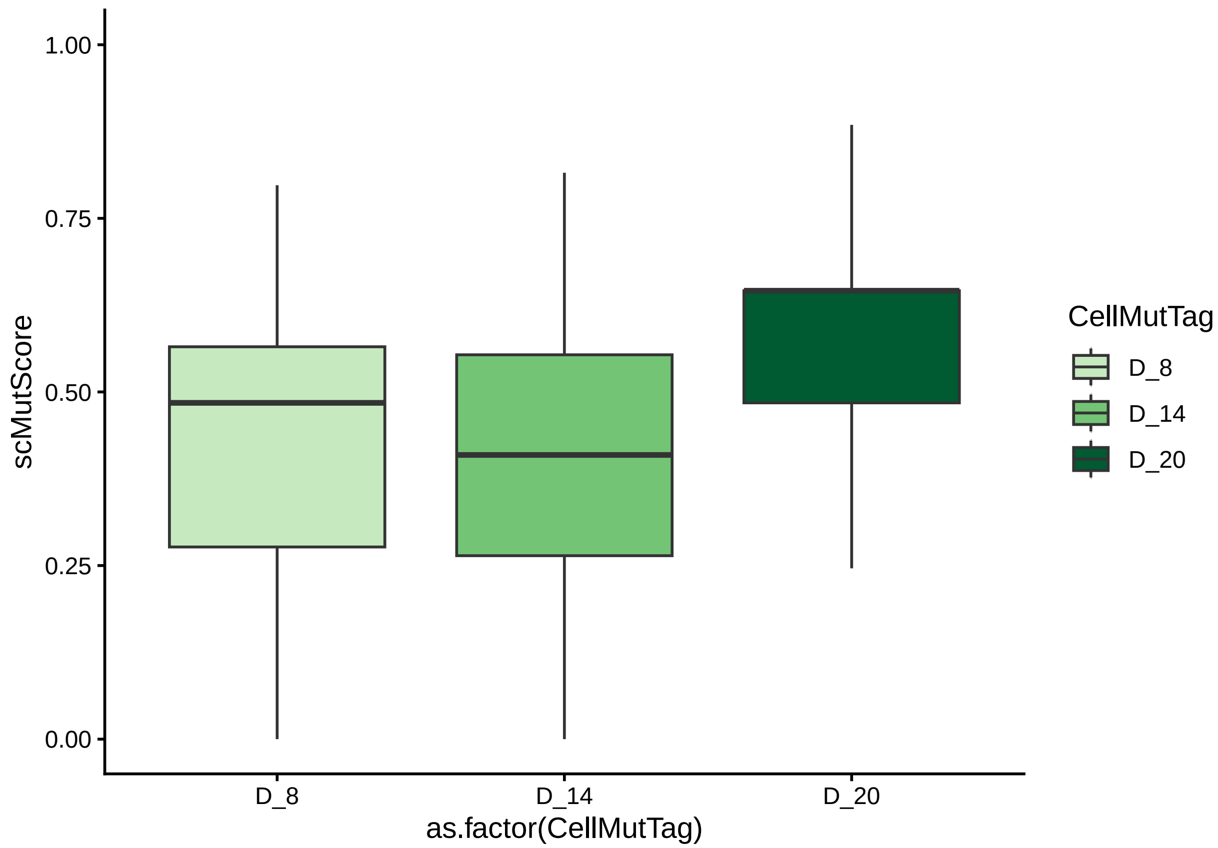
```r
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone),  alpha = 0.1,linetype='dashed') +
  scale_fill_manual(values = day_color) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

# EpiTraceAge
res_index %>% dplyr::group_by(clone,Group,cell_day) %>% dplyr::summarise(meanEpi=mean(EpiTraceAge_itera
## `summarise()` has grouped output by 'clone', 'Group'. You can override using
## the `.groups` argument.
res_index_meta_summary_EpiTraceAge$cell_group <- 'Prog'
res_index_meta_summary_EpiTraceAge$cell_group[grepl('my1',res_index_meta_summary_EpiTraceAge$Group)] <-
res_index_meta_summary_EpiTraceAge$cell_group[grepl('my2|my3|my4',res_index_meta_summary_EpiTraceAge$Gr
res_index_meta_summary_EpiTraceAge$cell_group <- factor(res_index_meta_summary_EpiTraceAge$cell_group,le

p_CloneLevelByCellGroup_EpiTraceAge <- ggplot(res_index_meta_summary_EpiTraceAge, mapping = aes(x = as.
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone),  alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "EpiTraceAge") +
  theme_classic()

p_CloneLevelByCellGroup_scMutraceScore <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = a
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone),  alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "EpiTraceAge") +
  theme_classic()

p_pseudotimeByDaysEry <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=pseudutime_ery, fill = CellMu
     scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
     theme_classic()

p_pseudotimeByDaysMy <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=pseudutime_my, fill = CellMutT
     scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
     theme_classic()

p_pseudotimeByGroupQunEry <- ggplot(res_index, aes(x=as.factor(GroupQun), y=pseudutime_ery, fill = Grou
     scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
     theme_classic()

p_pseudotimeByGroupQunMy <- ggplot(res_index, aes(x=as.factor(GroupQun), y=pseudutime_my, fill = GroupQu
     scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
     theme_classic()

p_UMAP_scMutrace <- ggplot(res_index, aes(x = x, y = y, fill =  as.numeric(scMutScore))) +
  geom_point(shape = 21, size = 2, alpha = 0.3) +
  scale_fill_gradientn(colours = c("#313695", "#4575b4", "#74add1", "#abd9e9", "#e0f3f8", "#ffffbf", "#
  theme_classic() + labs(fill = "scMutScore")

df_plot <- res_index %>%
  count(CellMutTag, GroupQun) %>%
  complete(
```

```r
    CellMutTag,
    GroupQun = c(
      "prog", "prog_my",
      "my1", "my2", "my3", "my4",
      "prog_ery", "ery1_2", "ery3_4", "ery5_6"
    ),
    fill = list(n = 0)
  )

df_plot$GroupQun <- factor(
  df_plot$GroupQun,
  levels = c(
    "prog", "prog_my",
    "my1", "my2", "my3", "my4",
    "prog_ery", "ery1_2", "ery3_4", "ery5_6"
  )
)

ggplot(df_plot, aes(x = CellMutTag, y = n, fill = GroupQun)) +
  geom_col(position = "dodge") +
  labs(
    x = "CellMutTag",
    y = "Cell count",
    fill = "GroupQun"
  ) +
  scale_fill_manual(values = cluster_color) +
  theme_classic() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

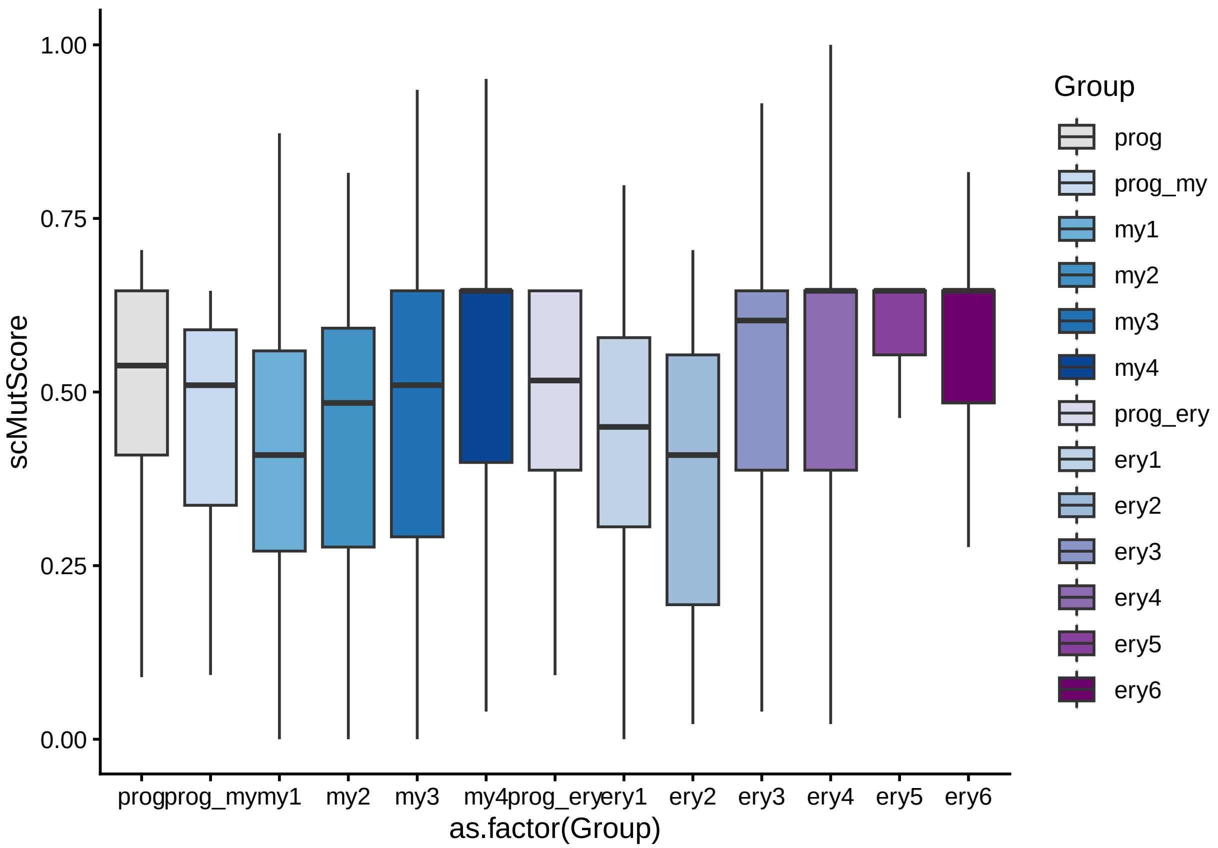



```
df_mean <- res_index %>%
  group_by(GroupQun) %>%
  summarise(
    mean_scMutScore = mean(scMutScore, na.rm = TRUE),
    mean_EpiTraceAge  = mean(EpiTraceAge_iterative, na.rm = TRUE),
    mean_pseudutime_my     = mean(pseudutime_my, na.rm = TRUE),
    mean_pseudutime_ery     = mean(pseudutime_ery, na.rm = TRUE),
    n = n()
  )

df_frac <- res_index %>%
  group_by(GroupQun, CellMutTag) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(GroupQun) %>%
  mutate(prop = n / sum(n))

df_frac_2 <- res_index %>%
  group_by(GroupQun, CellMutTag) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(CellMutTag) %>%
  mutate(prop = n / sum(n))

p_cellfrac <- ggplot(df_frac, aes(x = GroupQun, y = prop, fill = CellMutTag)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_fill_manual(values = day_color) +
```

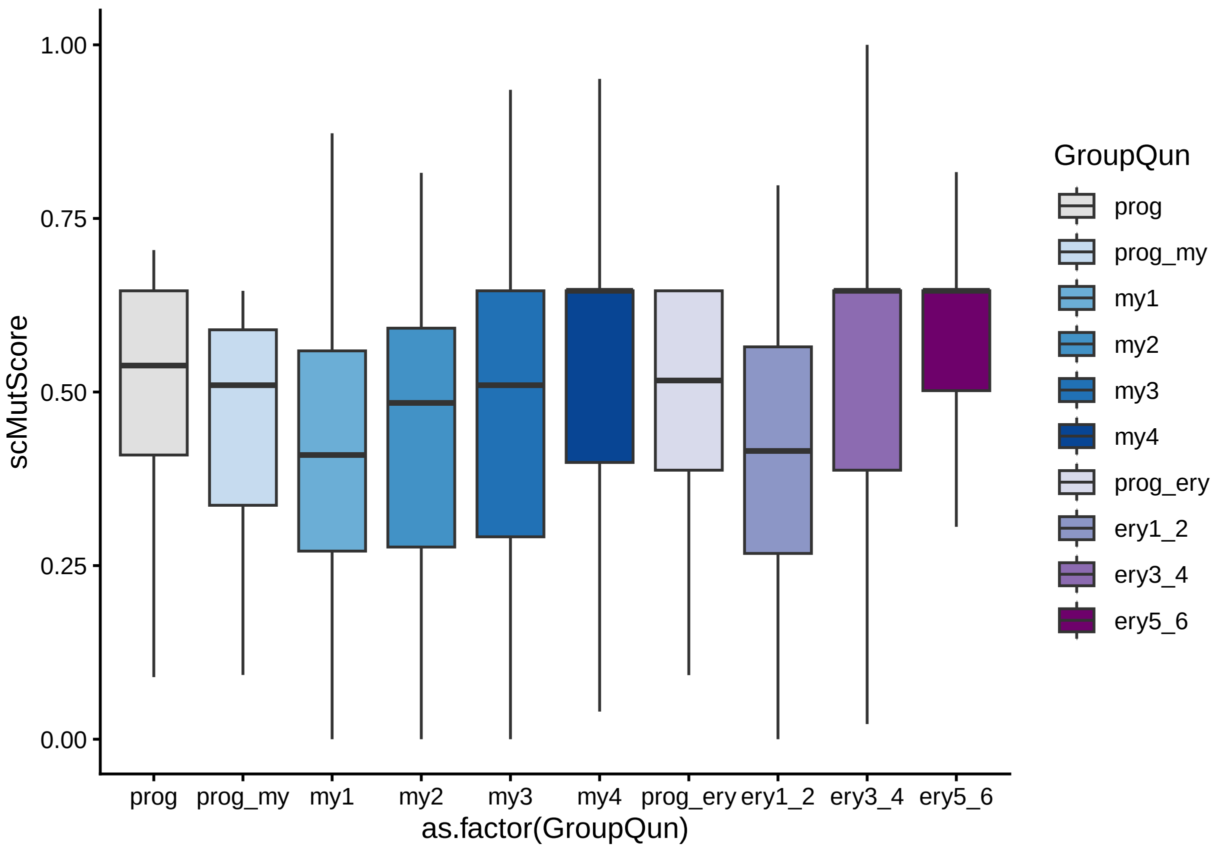
```r
  scale_y_continuous(labels = scales::percent) +
  theme_bw()

p_scMutScorebydays
```



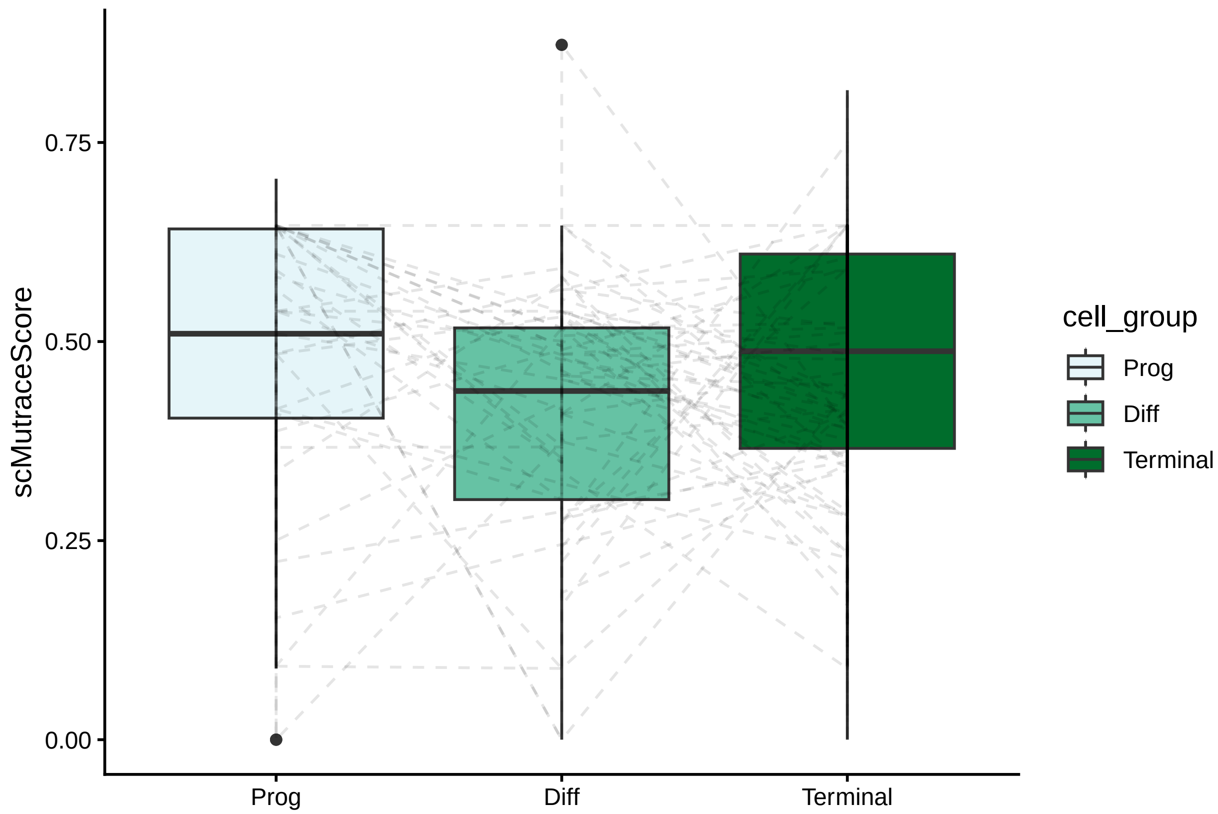

```r
p_scMutScorebycelltypes
```



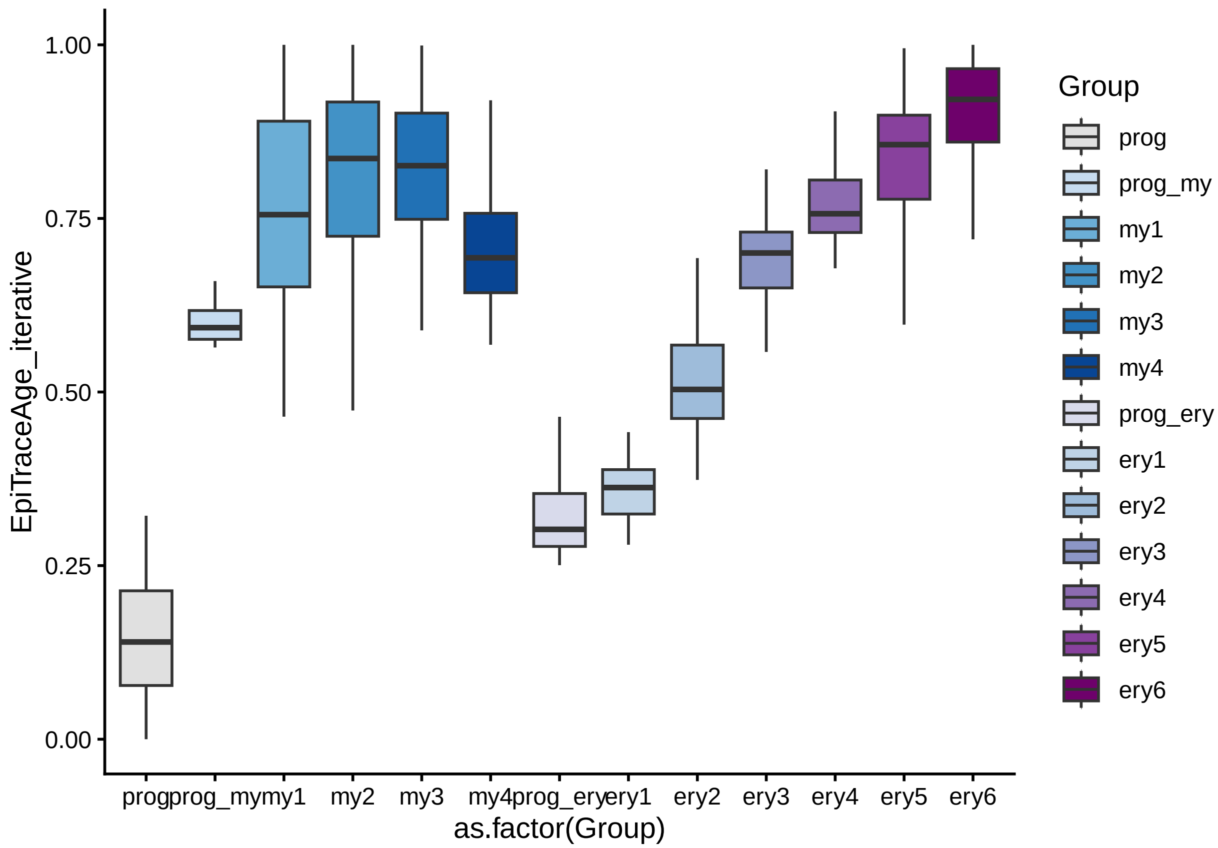

```
p_scMutScorebycelltypesQun
```



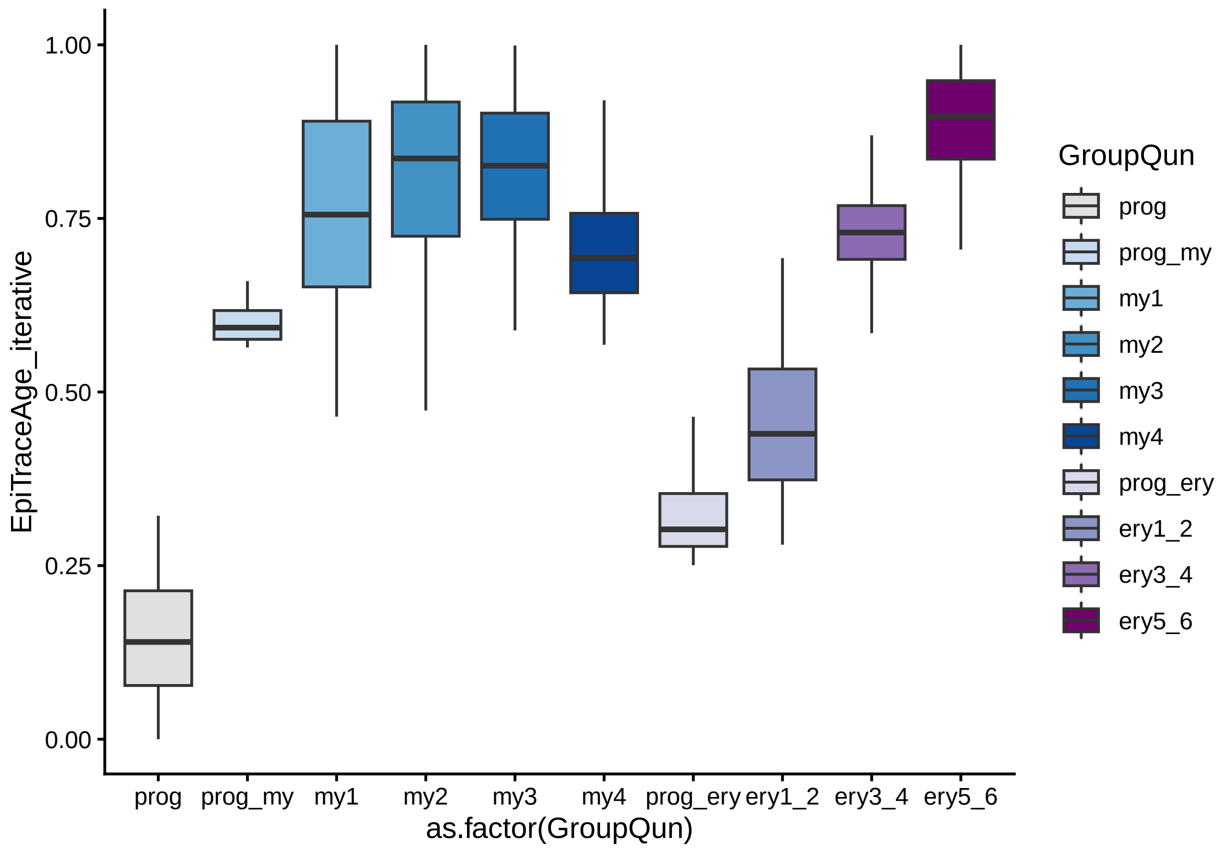

```
p_CloneLevelByCellGroup
```



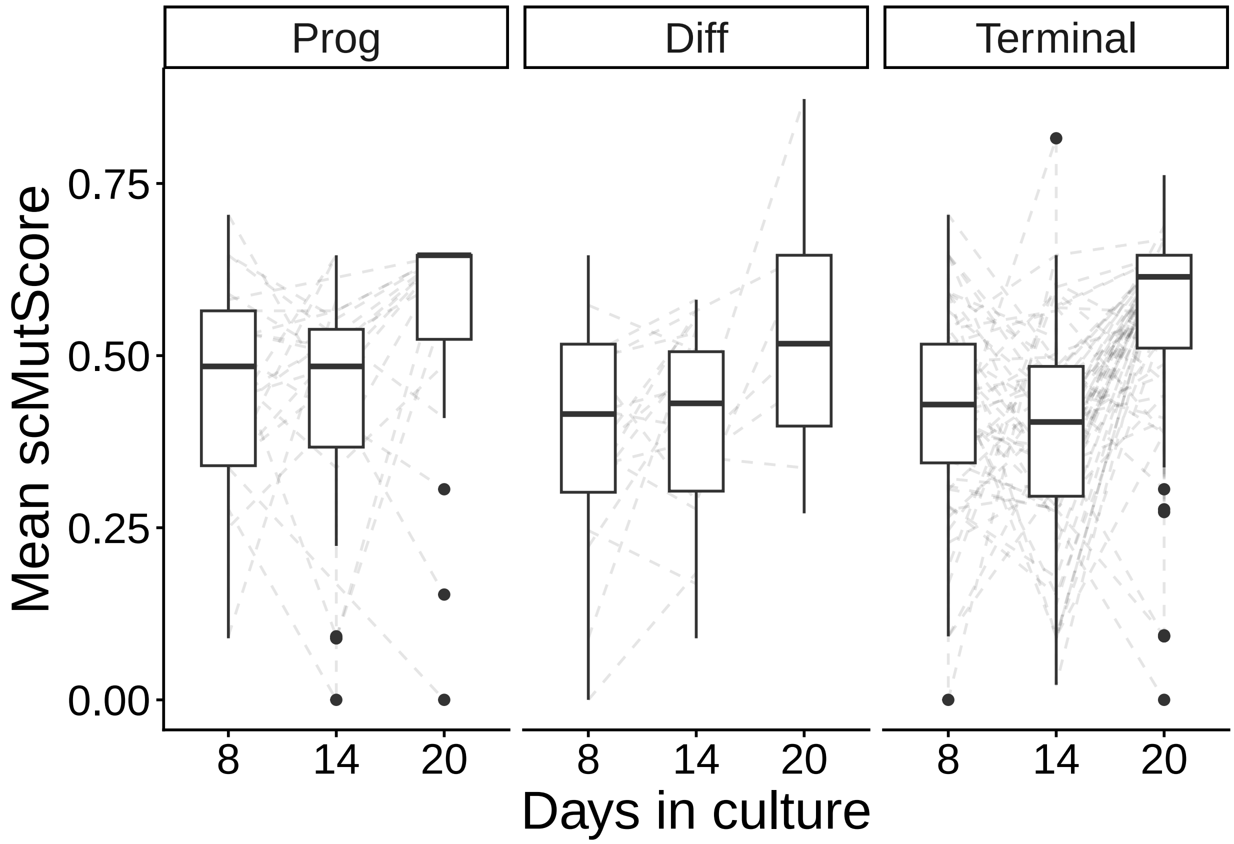

```
p_EpiScorebyCelltype
```



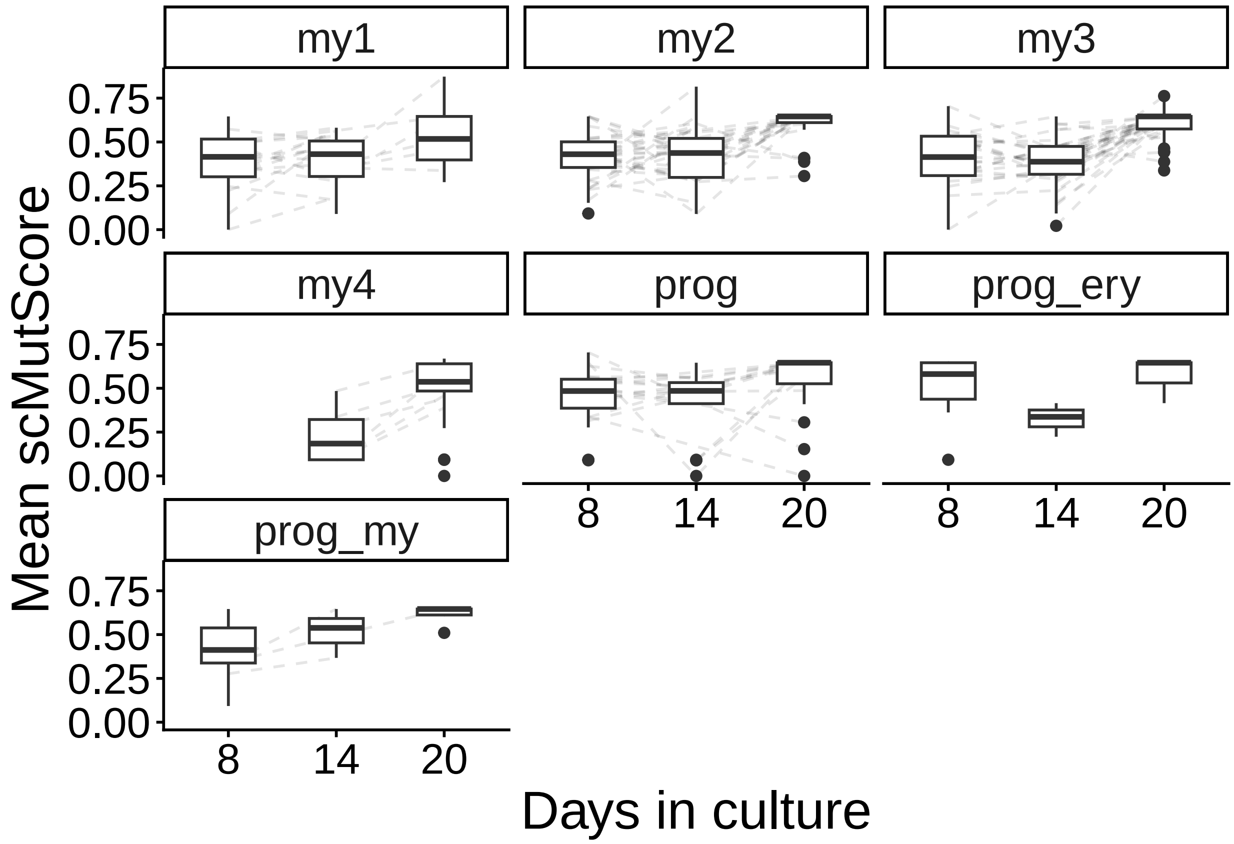

```
p_EpiScorebyCelltypeQun
```



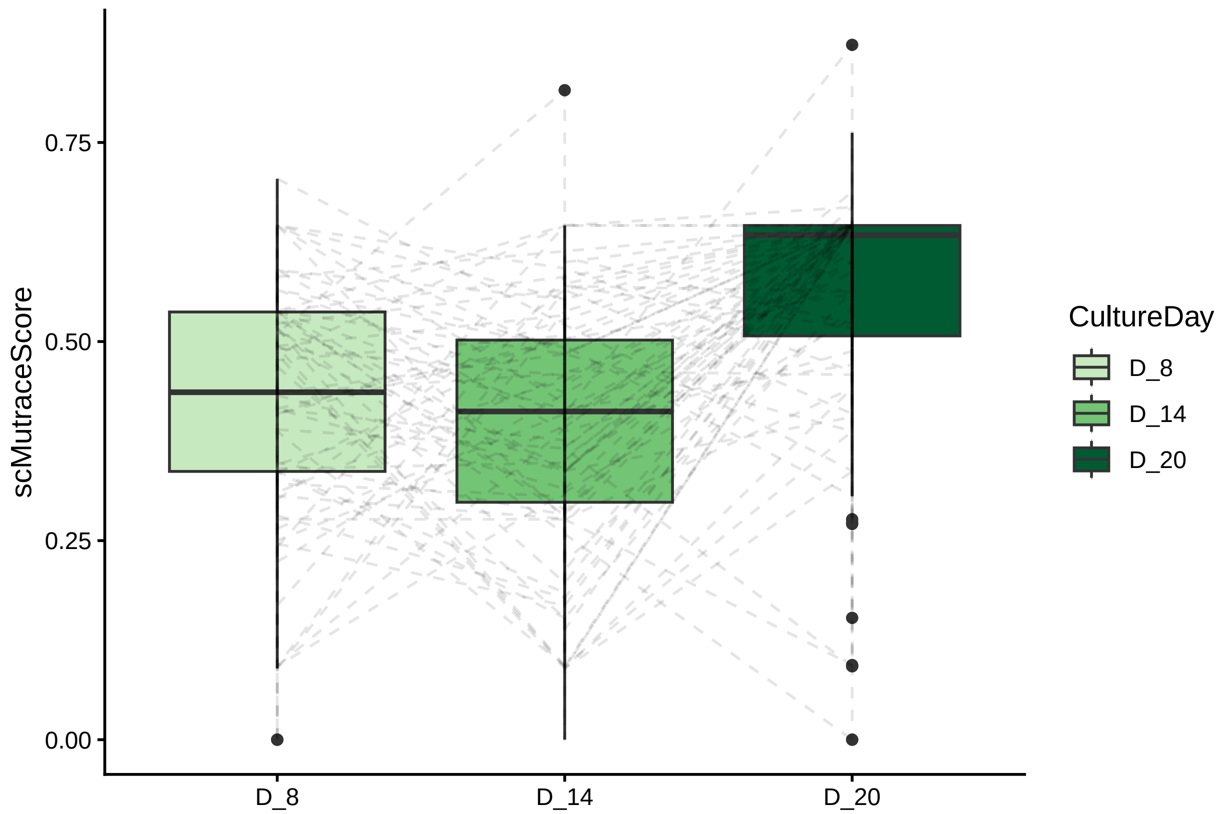

```
p_CloneLevelByDaysSplitbyCellgroup
```



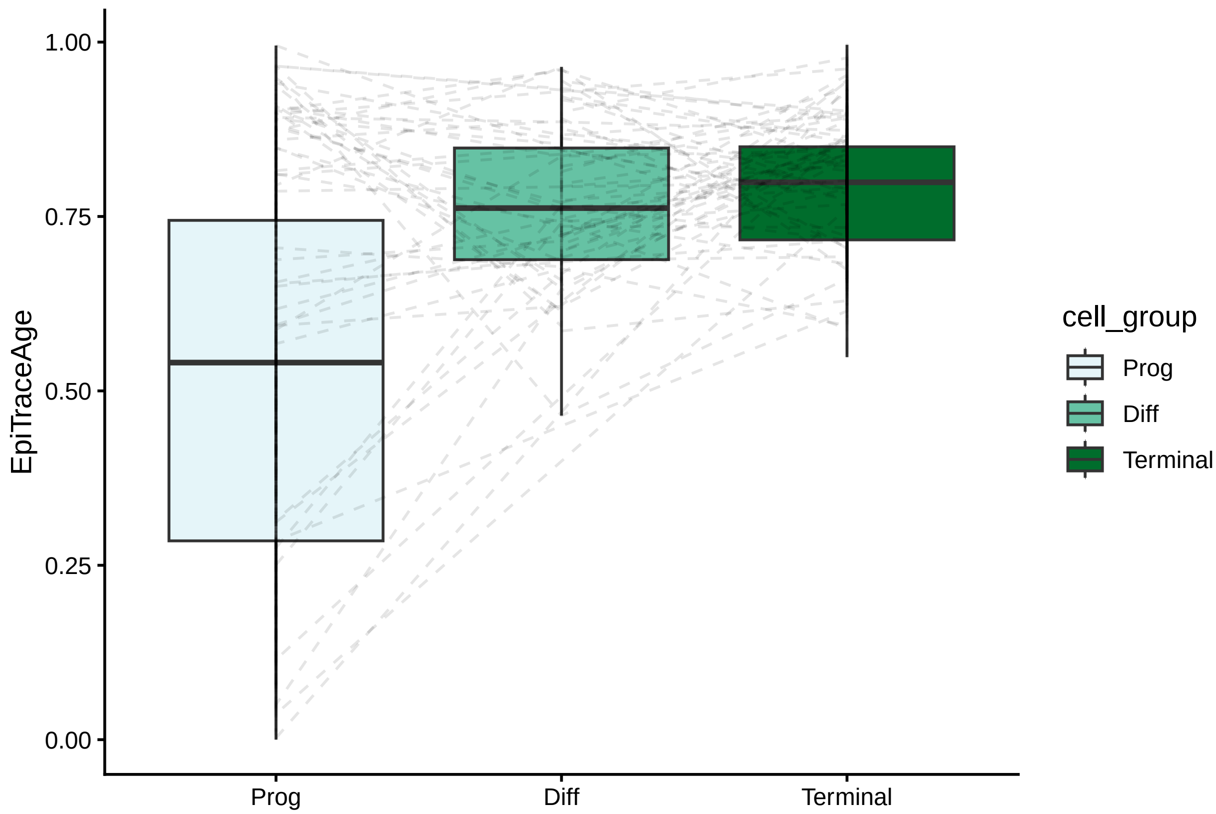

```
p_CloneLevelByDaysSplitbyCellType
## `geom_line()`: Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?
```



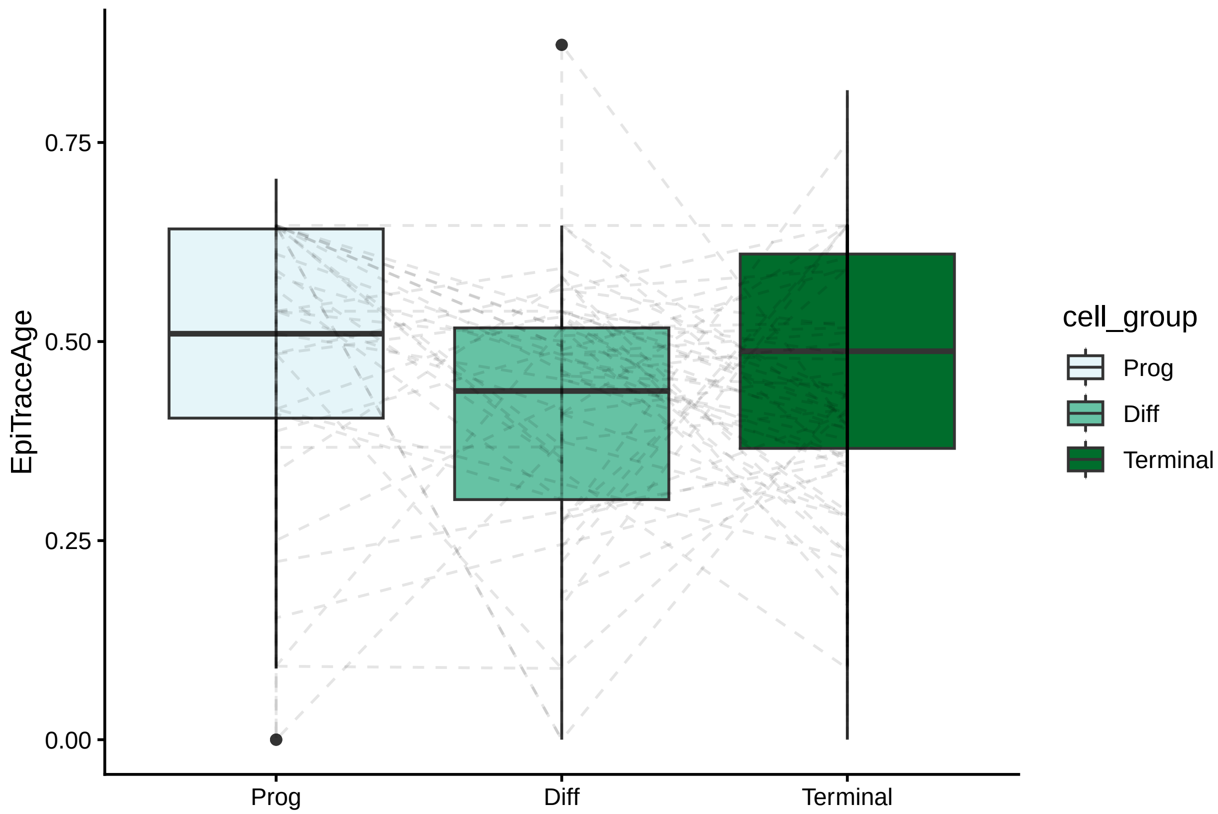

```
p_CloneLevelByDays
```



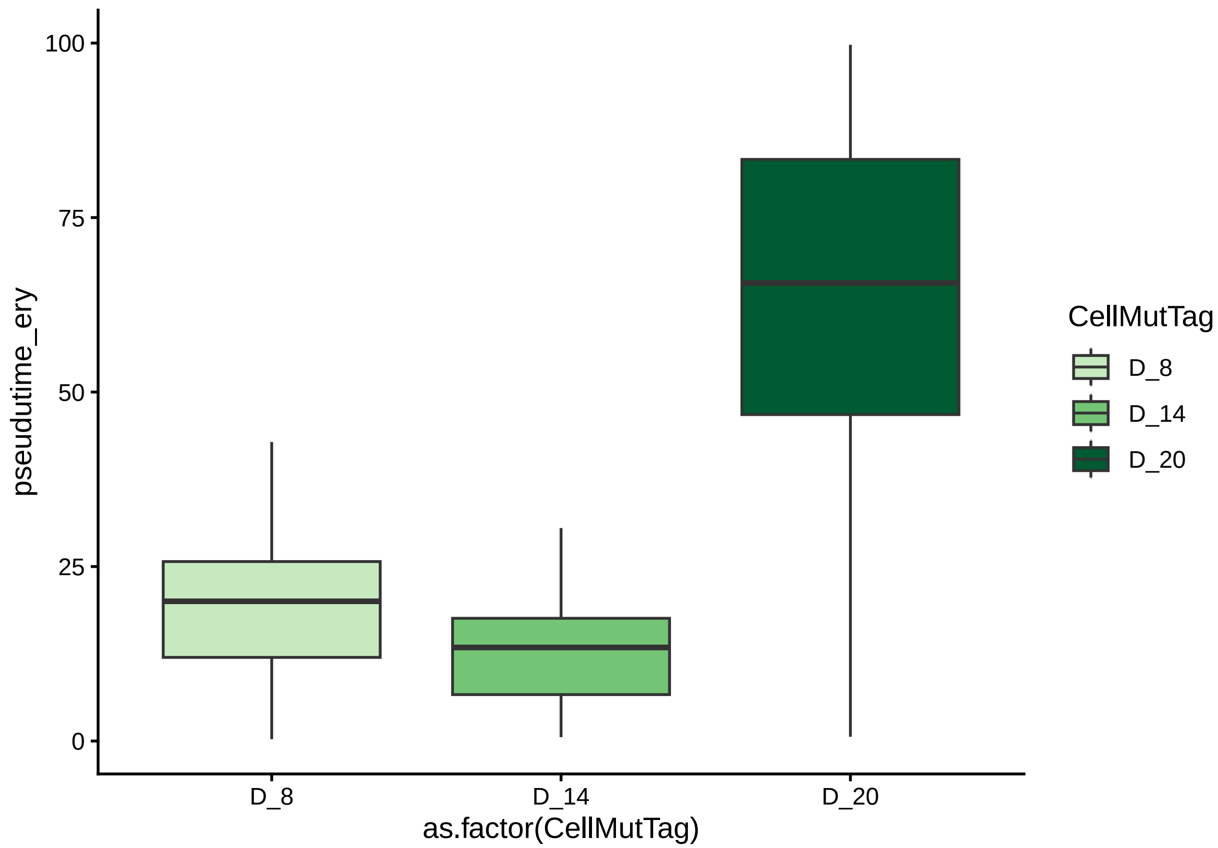

```
p_CloneLevelByCellGroup_EpiTraceAge
```



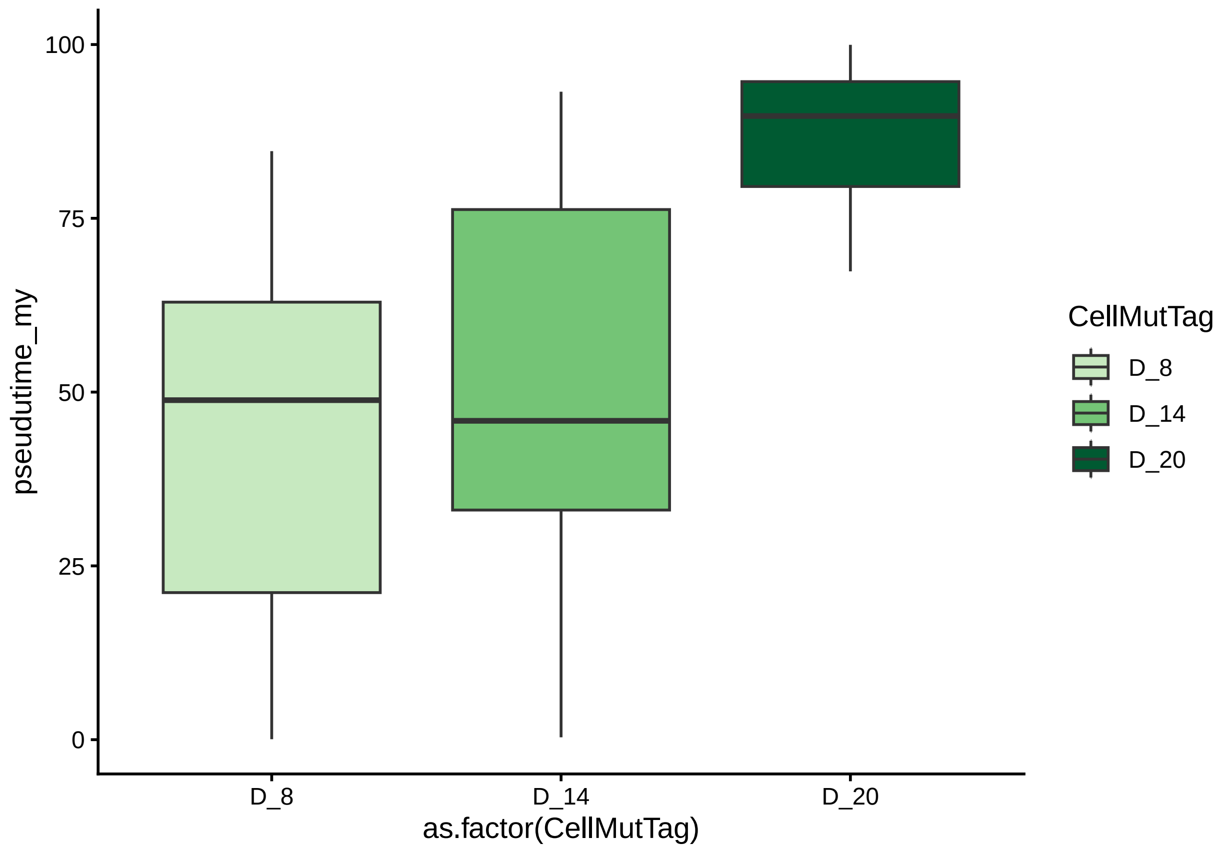

```
p_CloneLevelByCellGroup_scMutraceScore
```



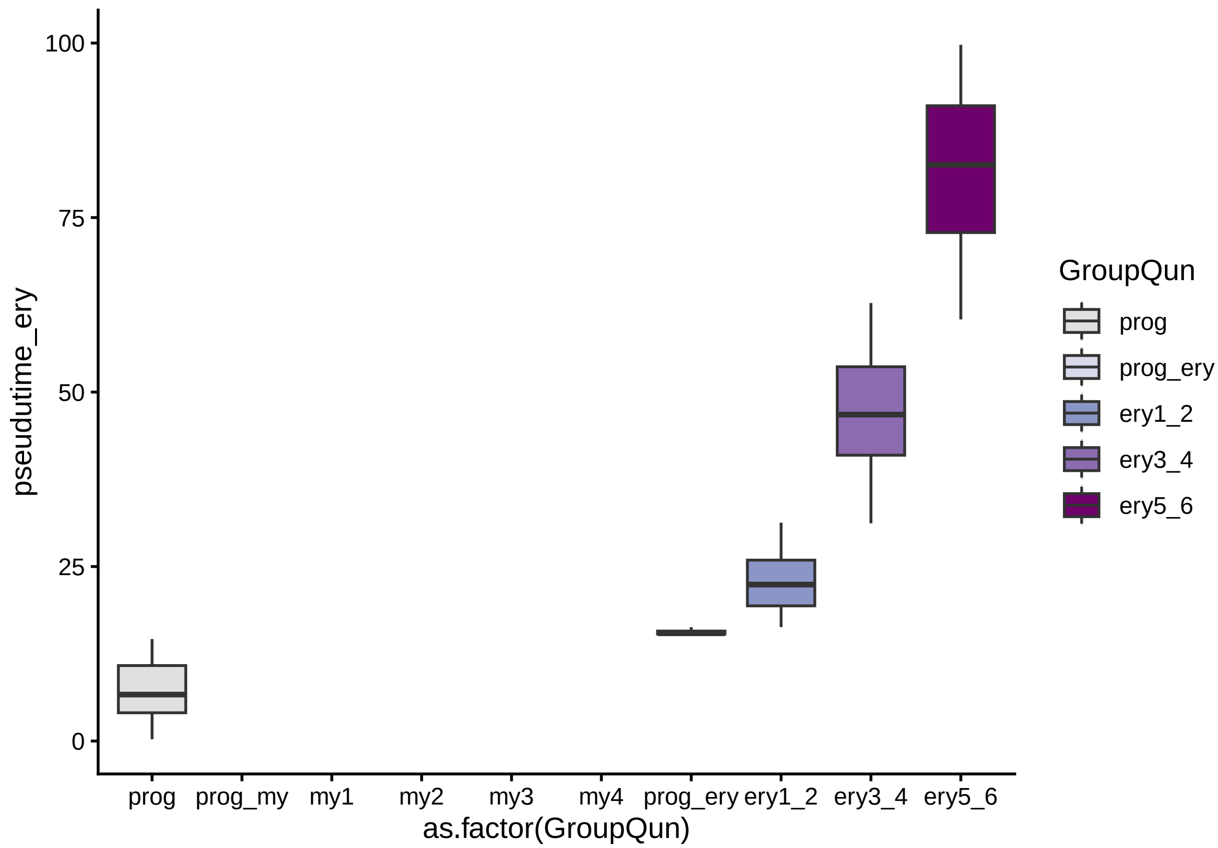

```
p_pseudotimeByDaysEry
## Warning: Removed 1252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



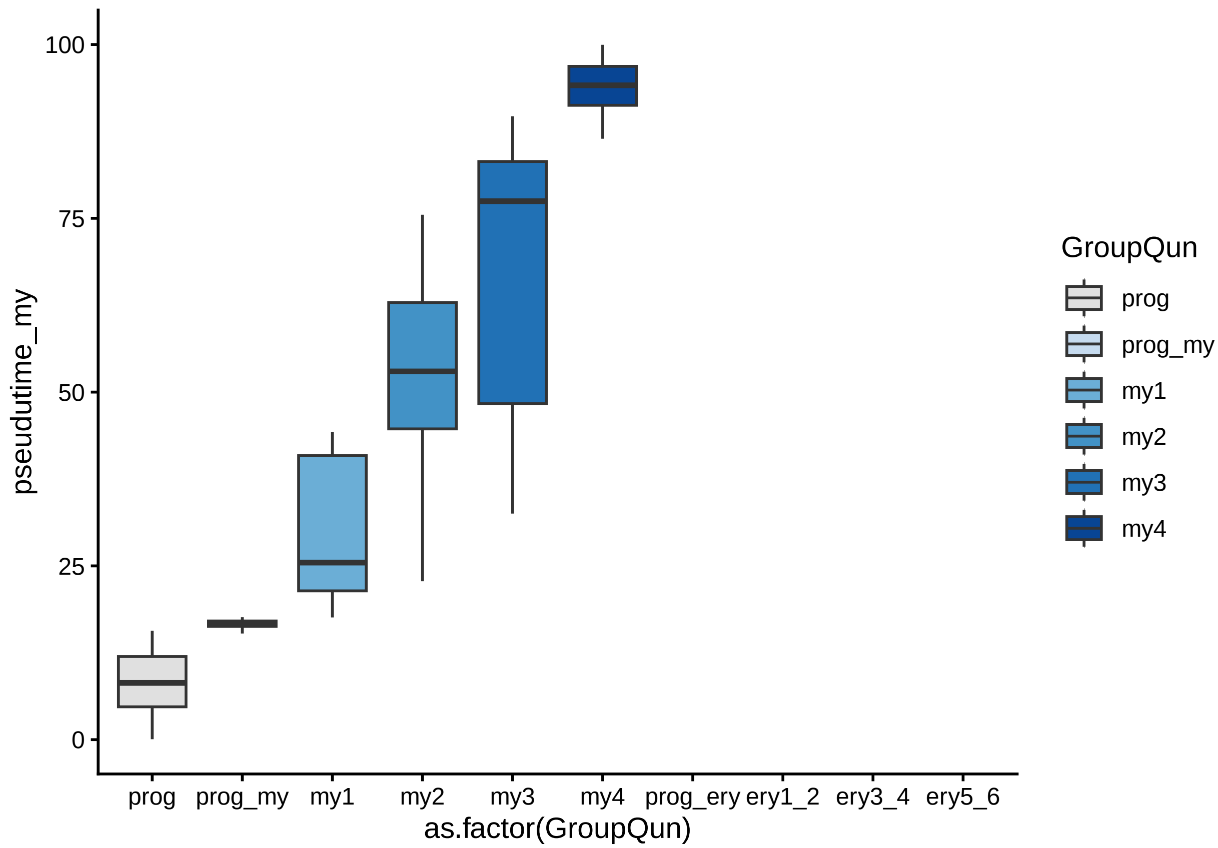

```
p_pseudotimeByDaysMy
## Warning: Removed 437 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



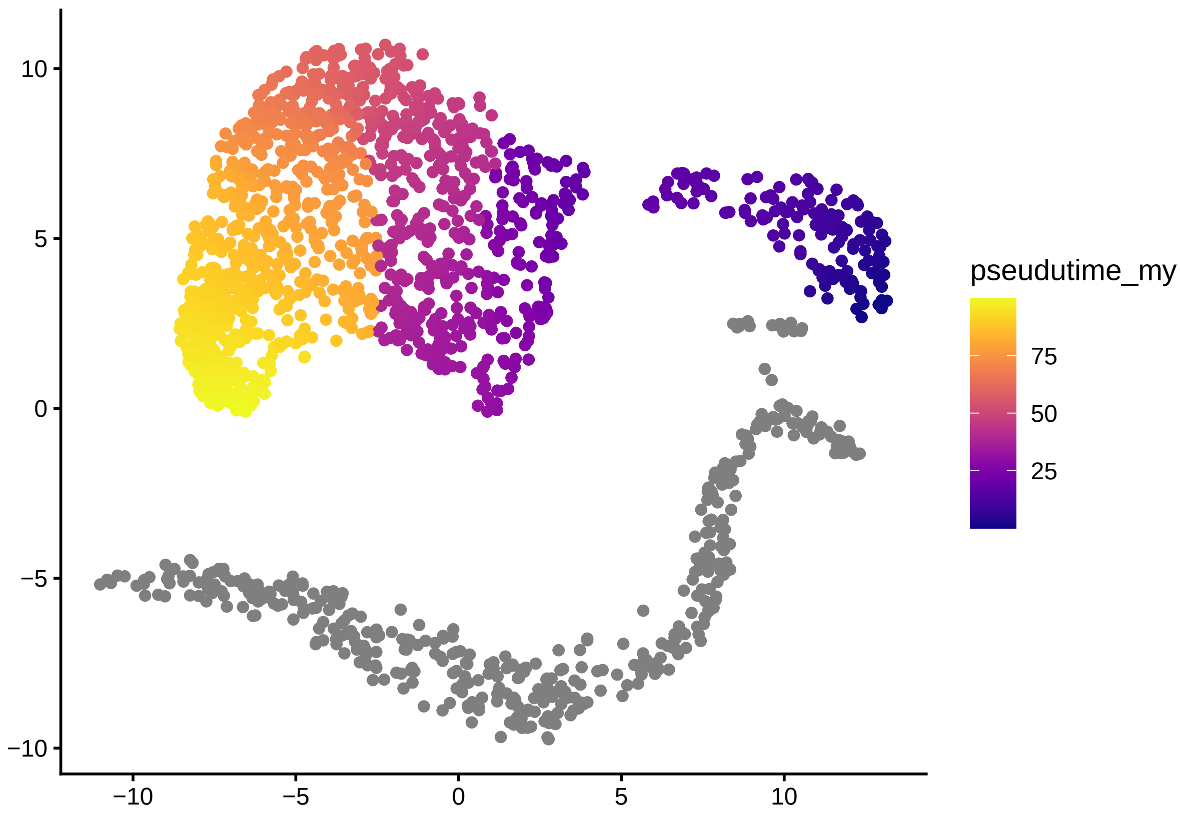

```
p_pseudotimeByGroupQunEry
## Warning: Removed 1252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



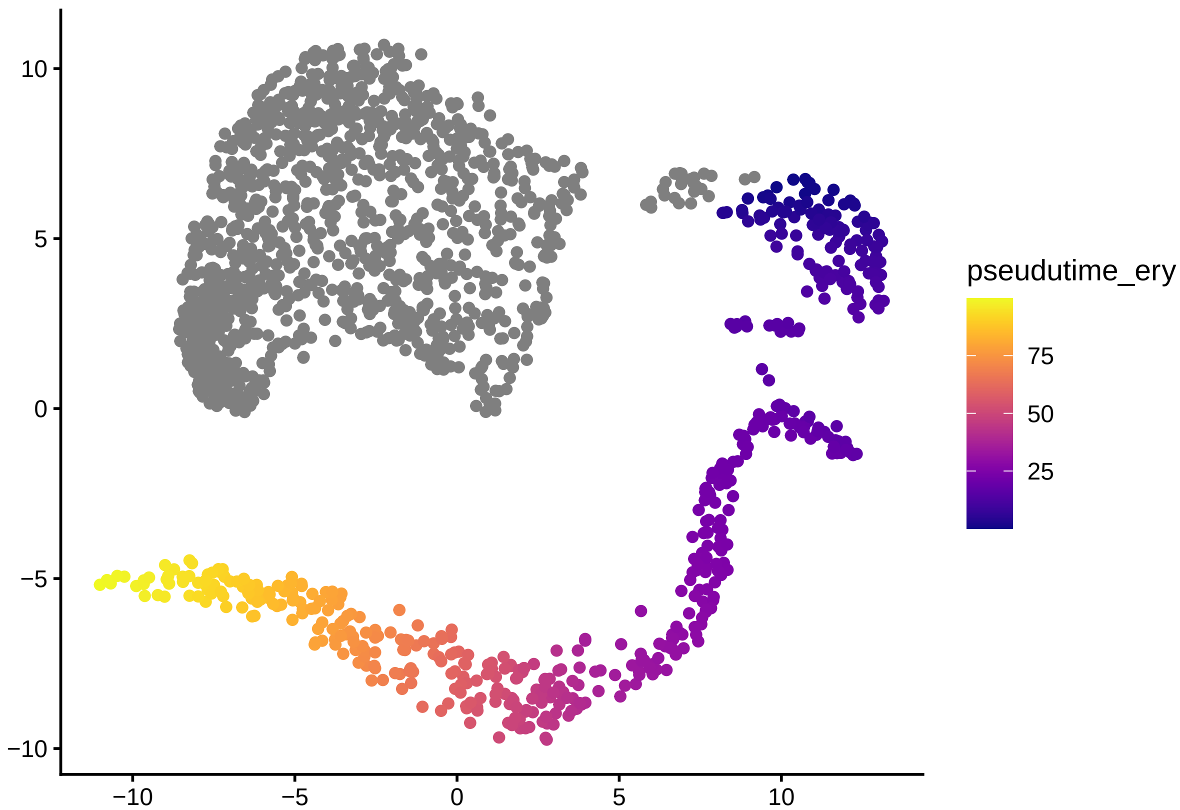

```
p_pseudotimeByGroupQunMy
## Warning: Removed 437 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



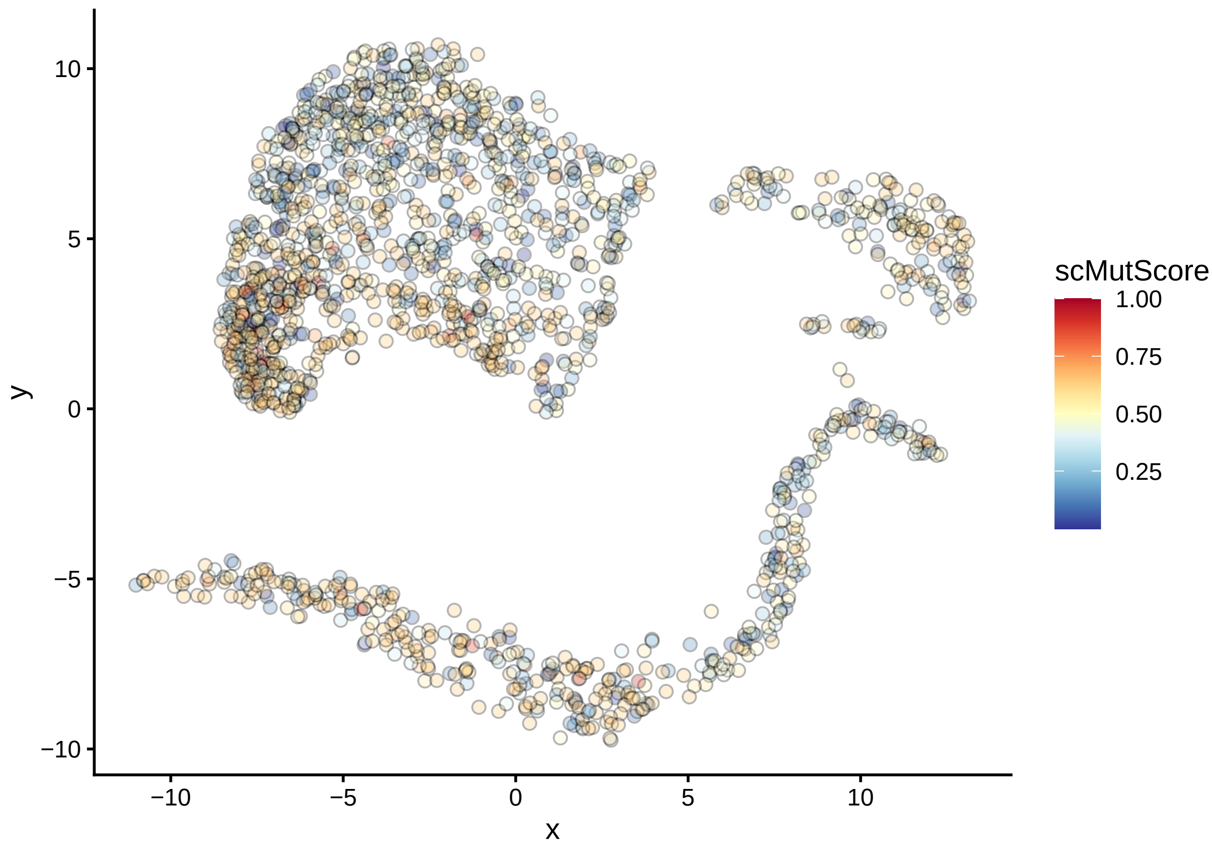

```
p_UMAP_pseudutime_my
```



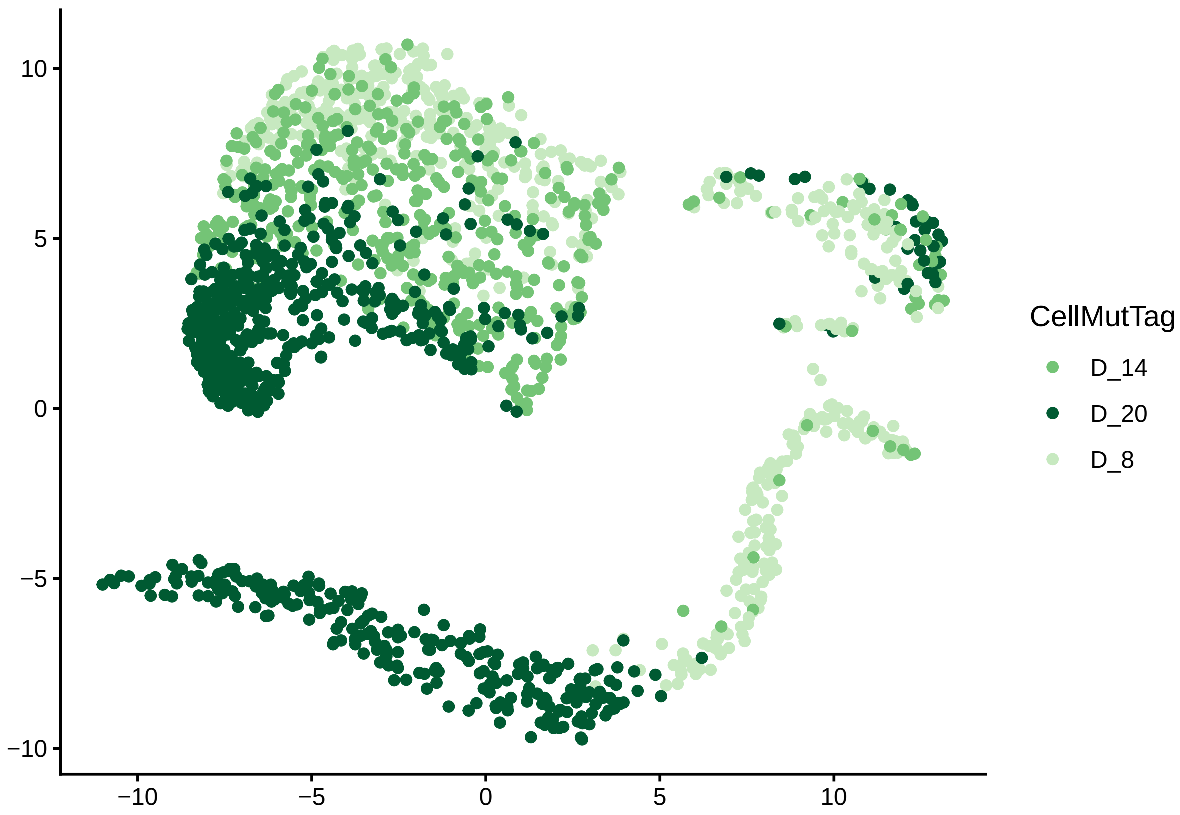

```
p_UMAP_pseudutime_ery
```



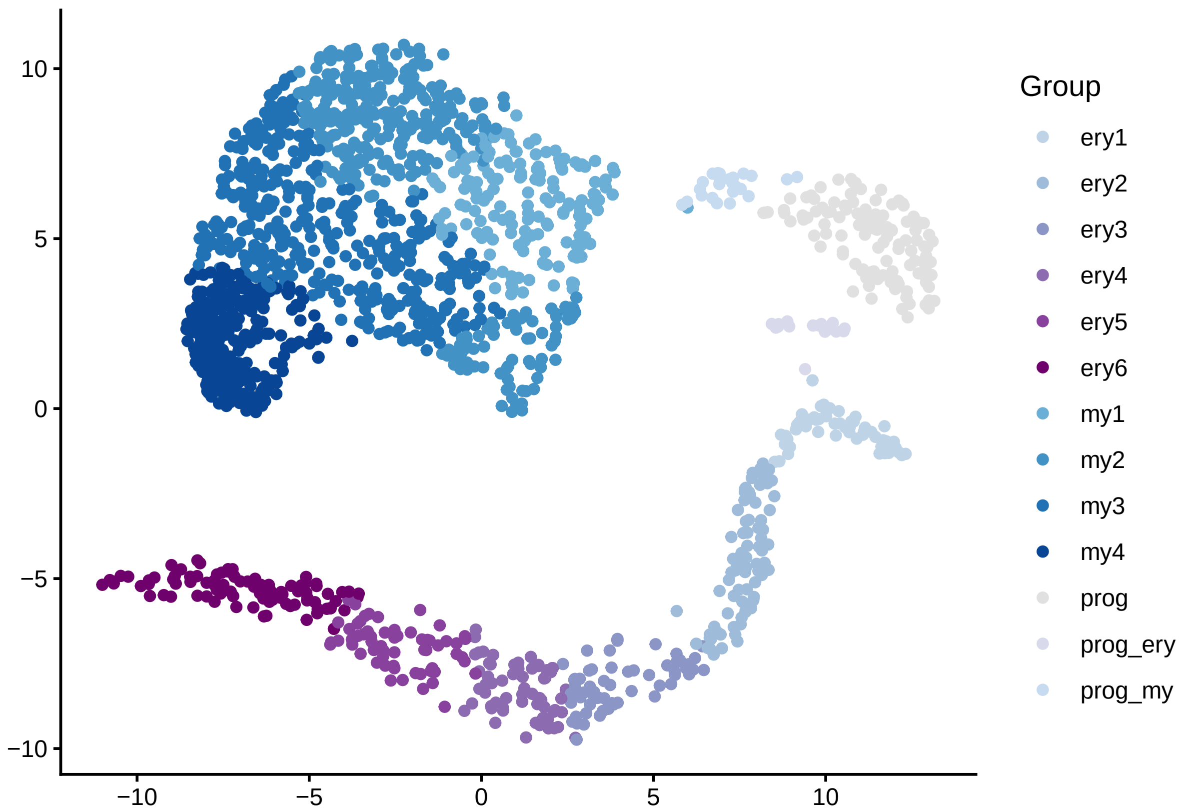

```
p_UMAP_scMutrace
```



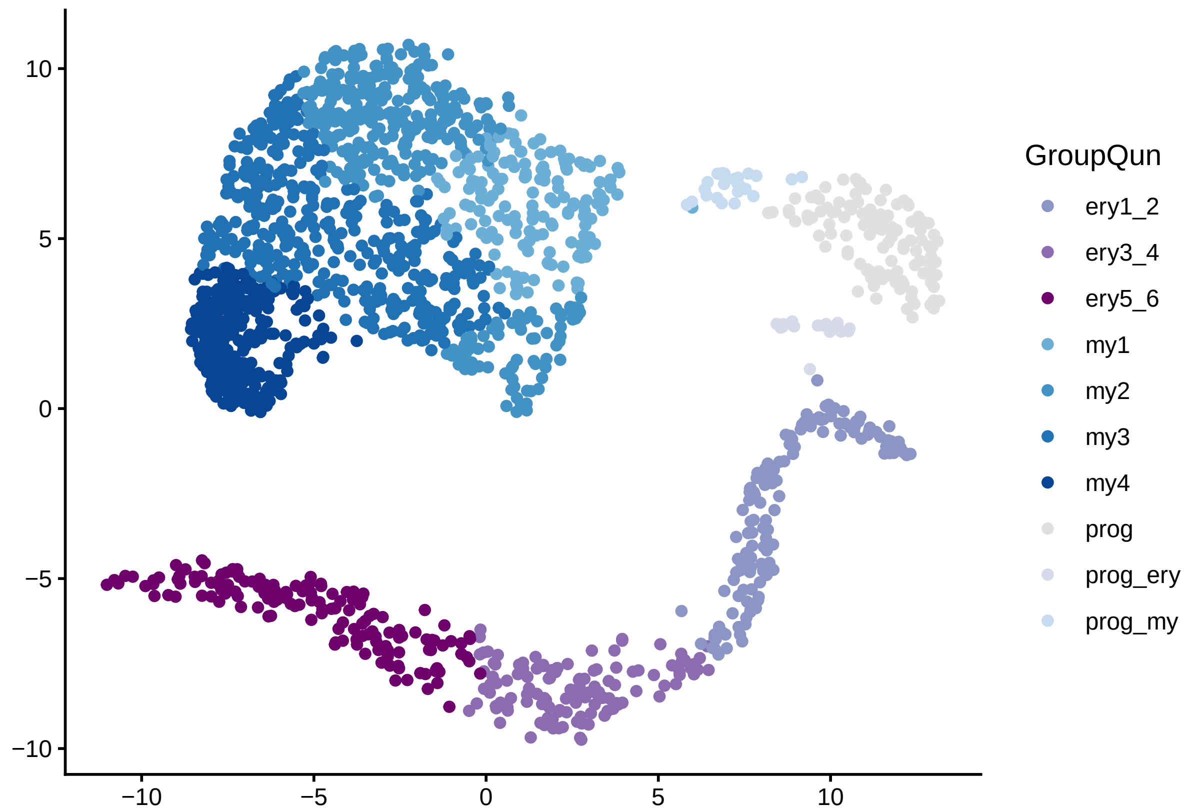

```
p_UMAP_Days
```




```
p_UMAP_Groups
```

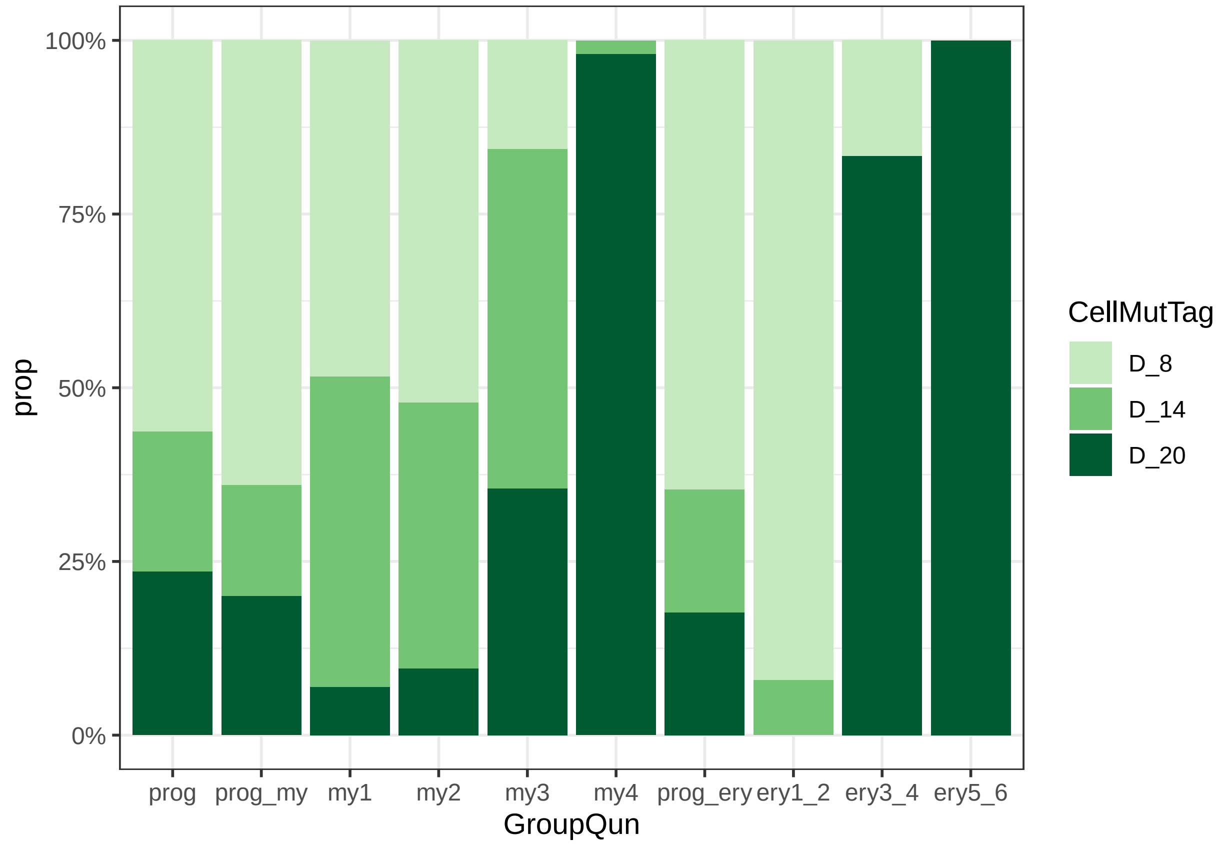



```
p_UMAP_GroupsQun
```



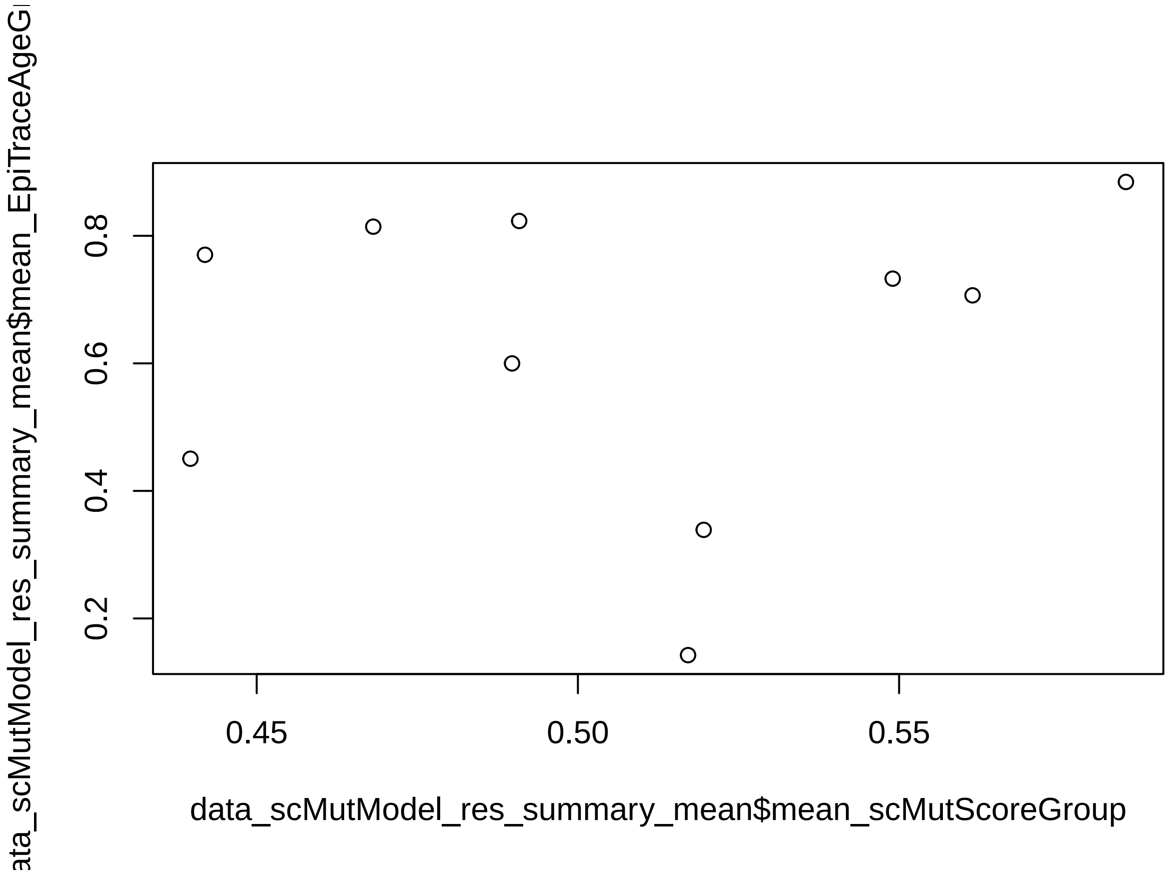
GroupQun
ery1_2
ery3_4
ery5_6
my1
my2
my3
my4
prog
prog_ery
prog_my
10
5
0
−5
−10
−10
−5
0
5
10


```
p_cellfrac
```

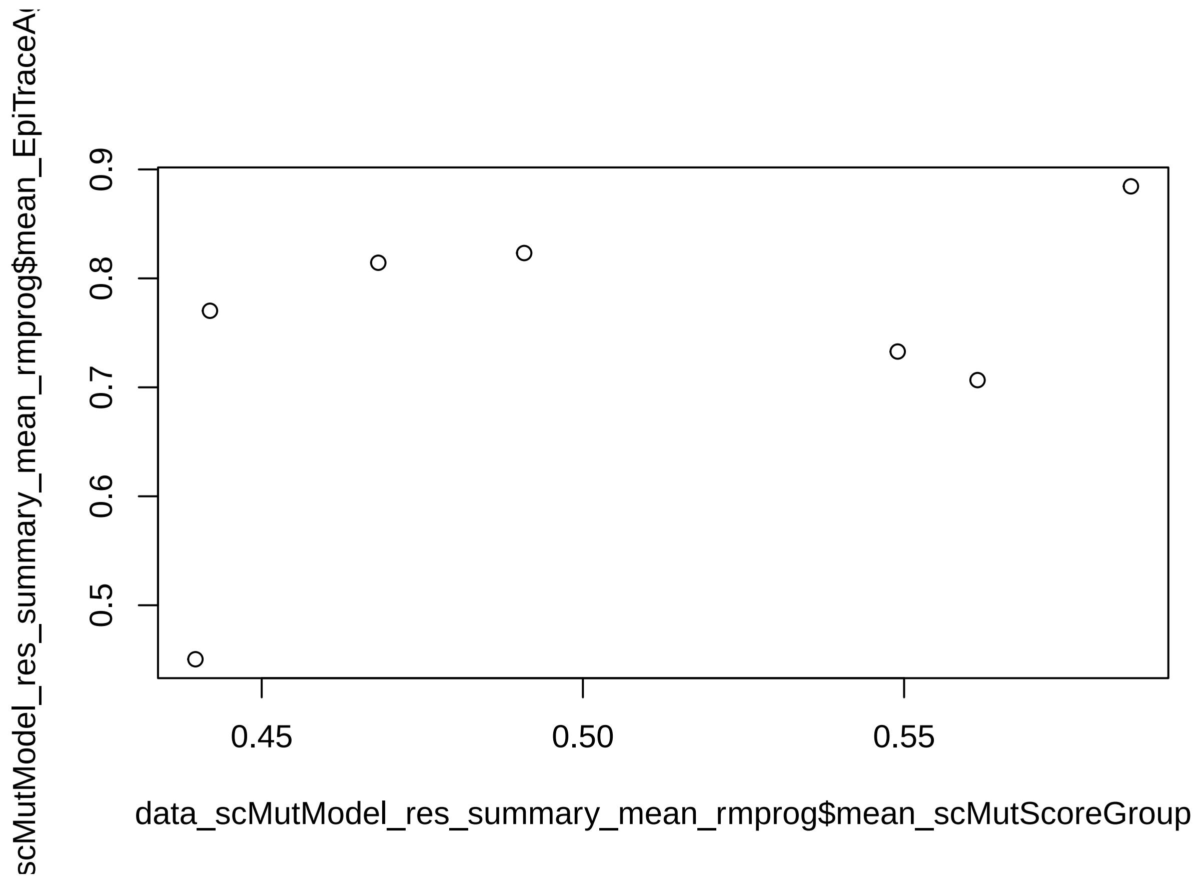


12 correlation of scMutrace and EpiAge

```
data_scMutModel_res_summary_mean <- data_scMutModel_res_summary %>%
  dplyr::group_by(GroupQun) %>%
  dplyr::summarise(mean_scMutScoreGroup = mean(mean_scMutScore, na.rm = TRUE), mean_EpiTraceAgeGroup = m
  dplyr::ungroup()

data_scMutModel_res_summary_mean <-
  data_scMutModel_res_summary_mean %>%
  mutate(
    lineage = case_when(
      GroupQun %in% c("prog", "prog_my", "prog_ery") ~ "prog",
      grepl("^my", GroupQun) ~ "my",
      grepl("^ery", GroupQun) ~ "ery",
      TRUE ~ NA_character_
    ))

data_scMutModel_res_summary_mean_rmprog <- data_scMutModel_res_summary_mean[!data_scMutModel_res_summary

plot(data_scMutModel_res_summary_mean$mean_scMutScoreGroup, data_scMutModel_res_summary_mean$mean_EpiTra
```




```
plot(data_scMutModel_res_summary_mean_rmprog$mean_scMutScoreGroup, data_scMutModel_res_summary_mean_rmp
```




```
cor.test(data_scMutModel_res_summary_mean$mean_scMutScoreGroup, data_scMutModel_res_summary_mean$mean_Ep
##
##  Pearson's product-moment correlation
##
## data:  data_scMutModel_res_summary_mean$mean_scMutScoreGroup and data_scMutModel_res_summary_mean$mea
## t = 0.34377, df = 8, p-value = 0.7399
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.5508178  0.6973067
## sample estimates:
##       cor
## 0.1206516
cor.test(data_scMutModel_res_summary_mean_rmprog$mean_scMutScoreGroup, data_scMutModel_res_summary_mean_
##
##  Pearson's product-moment correlation
##
## data:  data_scMutModel_res_summary_mean_rmprog$mean_scMutScoreGroup and data_scMutModel_res_summary_m
## t = 1.1128, df = 5, p-value = 0.3164
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.4627831  0.8974810
## sample estimates:
##       cor
## 0.4455505

cor_results <- data_scMutModel_res_summary_mean_rmprog %>%
```

```r
  group_by(lineage) %>%
  group_modify(~ broom::tidy(
    cor.test(
      .x$mean_scMutScoreGroup,
      .x$mean_EpiTraceAgeGroup,
      method = "pearson"
    )
  )) %>%
  ungroup()
cor_results
## # A tibble: 2 x 9
##   lineage estimate statistic p.value parameter method       alternative conf.low
##   <chr>      <dbl>     <dbl>   <dbl>     <int> <chr>        <chr>          <dbl>
## 1 ery        0.994      9.10  0.0697         1 Pearson's p~ two.sided     NA
## 2 my        -0.669     -1.27  0.331          2 Pearson's p~ two.sided     -0.992
## # i 1 more variable: conf.high <dbl>
```

13 savePic

```r
pdf("../Figure/CD34/p_UMAP_Days.pdf", width = 4.47, height = 3.20)
p_UMAP_Days
dev.off()

pdf("../Figure/CD34/p_UMAP_Groups.pdf", width = 4.47, height = 3.20)
p_UMAP_Groups
dev.off()

pdf("../Figure/CD34/p_UMAP_pseudutime_my.pdf", width = 4.47, height = 3.20)
p_UMAP_pseudutime_my
dev.off()

pdf("../Figure/CD34/p_UMAP_pseudutime_ery.pdf", width = 4.47, height = 3.20)
p_UMAP_pseudutime_ery
dev.off()

pdf("../Figure/CD34/p_scMutScorebydays.pdf", width = 4.47, height = 3.20)
p_scMutScorebydays
dev.off()

pdf("../Figure/CD34/p_scMutScorebycelltypes.pdf", width = 4.47, height = 3.20)
p_scMutScorebycelltypesQun
dev.off()

pdf("../Figure/CD34/p_EpiScorebyCelltype.pdf", width = 4.47, height = 3.20)
p_EpiScorebyCelltypeQun
dev.off()

pdf("../Figure/CD34/p_pseudotimeByDaysEry.pdf", width = 4.47, height = 3.20)
p_pseudotimeByDaysEry
dev.off()
```

```r
pdf("../Figure/CD34/p_pseudotimeByDaysMy.pdf", width = 4.47, height = 3.20)
p_pseudotimeByDaysMy
dev.off()

pdf("../Figure/CD34/p_pseudotimeByGroupQunEry.pdf", width = 4.47, height = 3.20)
p_pseudotimeByGroupQunEry
dev.off()

pdf("../Figure/CD34/p_pseudotimeByGroupQunMy.pdf", width = 4.47, height = 3.20)
p_pseudotimeByGroupQunMy
dev.off()

pdf("../Figure/CD34/p_cellfracbylineage.pdf", width = 4.47, height = 3.20)
p_cellfrac
dev.off()

pdf("../Figure/CD34/p_scMutrace_iter_Day.pdf", width = 4.47, height = 3.20)
p_scMutrace_iter_Day_scaled
dev.off()

pdf("../Figure/CD34/p_scMutrace_iter_GroupQun.pdf", width = 4.47, height = 3.20)
p_scMutrace_iter_GroupQunScaled
dev.off()

pdf("../Figure/CD34/p_iter_cellfrac.pdf", width = 4.47, height = 3.20)
p_iter_cellfrac
dev.off()

pdf("../Figure/CD34/p_randomMutNum_est.pdf", width = 4.47, height = 3.20)
p_randomMutNum_est
dev.off()
```

14 sessionInfo

```r
sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
```

```
##
## other attached packages:
##  [1] patchwork_1.3.2    lubridate_1.9.4    forcats_1.0.0      purrr_1.1.0
##  [5] readr_2.1.5        tibble_3.3.0       tidyverse_2.0.0    networkD3_0.4.1
##  [9] tidygraph_1.3.1    reshape2_1.4.4     pheatmap_1.0.13    corrplot_0.95
## [13] stringr_1.5.2      ggpmisc_0.6.2      ggpp_0.5.9         webshot_0.5.5
## [17] ggraph_2.2.2       igraph_2.1.4       Seurat_5.3.0       SeuratObject_5.2.0
## [21] sp_2.2-0           ggpubr_0.6.1       ggplot2_4.0.0      tidyr_1.3.1
## [25] dplyr_1.1.4        vegan_2.7-1        permute_0.9-8      ineq_0.2-13
## [29] FSA_0.10.0
##
## loaded via a namespace (and not attached):
##   [1] RcppAnnoy_0.0.22       splines_4.5.1          later_1.4.4
##   [4] polyclip_1.10-7        fastDummies_1.7.5      lifecycle_1.0.4
##   [7] rstatix_0.7.2          globals_0.18.0         lattice_0.22-7
##  [10] MASS_7.3-65            backports_1.5.0        magrittr_2.0.4
##  [13] plotly_4.11.0          rmarkdown_2.30         yaml_2.3.10
##  [16] httpuv_1.6.16          sctransform_0.4.2      spam_2.11-1
##  [19] spatstat.sparse_3.1-0  reticulate_1.43.0      cowplot_1.2.0
##  [22] pbapply_1.7-4          RColorBrewer_1.1-3     abind_1.4-8
##  [25] Rtsne_0.17             tweenr_2.0.3           data.tree_1.2.0
##  [28] ggrepel_0.9.6          irlba_2.3.5.1          listenv_0.9.1
##  [31] spatstat.utils_3.2-0   MatrixModels_0.5-4     goftest_1.2-3
##  [34] RSpectra_0.16-2        spatstat.random_3.4-2  fitdistrplus_1.2-4
##  [37] parallelly_1.45.1      codetools_0.2-20       ggforce_0.5.0
##  [40] tidyselect_1.2.1       farver_2.1.2           viridis_0.6.5
##  [43] matrixStats_1.5.0      spatstat.explore_3.5-3 jsonlite_2.0.0
##  [46] progressr_0.16.0       Formula_1.2-5          ggridges_0.5.7
##  [49] survival_3.8-3         tools_4.5.1            ica_1.0-3
##  [52] Rcpp_1.1.0             glue_1.8.0             gridExtra_2.3
##  [55] xfun_0.53              mgcv_1.9-3             withr_3.0.2
##  [58] fastmap_1.2.0          SparseM_1.84-2         digest_0.6.37
##  [61] timechange_0.3.0       R6_2.6.1               mime_0.13
##  [64] colorspace_2.1-2       scattermore_1.2        tensor_1.5.1
##  [67] dichromat_2.0-0.1      spatstat.data_3.1-8    utf8_1.2.6
##  [70] generics_0.1.4         data.table_1.17.8      graphlayouts_1.2.2
##  [73] httr_1.4.7             htmlwidgets_1.6.4      uwot_0.2.3
##  [76] pkgconfig_2.0.3        gtable_0.3.6           lmtest_0.9-40
##  [79] S7_0.2.0               htmltools_0.5.8.1      carData_3.0-5
##  [82] dotCall64_1.2          scales_1.4.0           png_0.1-8
##  [85] spatstat.univar_3.1-4  knitr_1.50             rstudioapi_0.17.1
##  [88] tzdb_0.5.0             nlme_3.1-168           zoo_1.8-14
##  [91] cachem_1.1.0           KernSmooth_2.23-26     parallel_4.5.1
##  [94] miniUI_0.1.2           pillar_1.11.1          grid_4.5.1
##  [97] vctrs_0.6.5            RANN_2.6.2             promises_1.3.3
## [100] car_3.1-3              xtable_1.8-4           cluster_2.1.8.1
## [103] evaluate_1.0.5         cli_3.6.5              compiler_4.5.1
## [106] crayon_1.5.3           rlang_1.1.6            future.apply_1.20.0
## [109] ggsignif_0.6.4         labeling_0.4.3         plyr_1.8.9
## [112] stringi_1.8.7          viridisLite_0.4.2      deldir_2.0-4
## [115] lazyeval_0.2.2         spatstat.geom_3.6-0    quantreg_6.1
## [118] Matrix_1.7-4           RcppHNSW_0.6.0         hms_1.1.3
## [121] future_1.67.0          shiny_1.11.1           ROCR_1.0-11
```

```
## [124] broom_1.0.10          memoise_2.0.1          polynom_1.4-1
```